

The allegory axioms, and what defines them in the Frobenius calculus

§1. Notation

Traditionally Adjunction $F \dashv G$ is defined as $F(x) \leq y$ iff $x \leq G(y)$, and you derive laws like this:

$$\begin{aligned} F(x) \leq F(x) &\implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x))) \\ G(y) \leq G(y) &\implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y) \end{aligned}$$

By define $X : * \mapsto x$, $Y : * \mapsto y$ and use diagram order, application can be replaced by composition: $XF \leq Y$ iff $X \leq YG$

• write F as $/$, G as $\backslash$

• render $a \leq b$ as $\boxed{\begin{smallmatrix} a \\ b \end{smallmatrix}}$

• **rewrite** $X \leq XF$ as $1 \leq F$

(1a)

$$\boxed{\frac{X/}{Y}} = \boxed{\frac{X}{Y\backslash}}$$

adj

$$\begin{aligned} \boxed{\frac{X/}{X/}} &\xRightarrow{\text{adj}} \boxed{\frac{X}{X\backslash}} \xRightarrow{\text{rewrite}} \boxed{\frac{*}{\bigwedge}} \xRightarrow[\text{unit}]{\text{put } \backslash \text{ on left}} \boxed{\frac{\backslash}{\bigwedge}} \\ &\xRightarrow[\text{put } / \text{ on right}]{\text{put } / \text{ on right}} \boxed{\frac{/}{\bigvee}} \\ \boxed{\frac{Y\backslash}{Y\backslash}} &\xRightarrow{\text{adj}} \boxed{\frac{Y\bigvee}{Y}} \xRightarrow{\text{rewrite}} \boxed{\frac{\bigvee}{*}} \xRightarrow[\text{counit}]{\text{put } / \text{ on left}} \boxed{\frac{\bigvee}{/}} \\ &\xRightarrow[\text{put } \backslash \text{ on right}]{\text{put } \backslash \text{ on right}} \boxed{\frac{\bigwedge}{\backslash}} \end{aligned}$$

now we have $\bigvee = /$ and $\bigwedge = \backslash$, and you just discovered string diagrams.

§1.1. Adjunctions

(1.1a)

$F \dashv G$	F's type	monad FG	$1 \in FG$	$GF \in 1$	F preserves u	G preserves n	$FGF=F$	$GFG=G$
$\triangleleft \dashv \triangle$	$A \rightarrow A \otimes A$	$\triangleleft \triangleright = 1$	$1 \in \triangleleft \triangleright$	$\triangleright \triangleleft \in 1$	$(RuS) \triangleleft = R \triangleleft uS \triangleleft$	$(RnS) \triangleright = R \triangleright nS \triangleright$	$\triangleleft \triangleright \triangleleft = \triangleleft$	$\triangleright \triangleleft \triangleright = \triangleright$
$\multimap \dashv \multimap$	$A \rightarrow I$	$\multimap \multimap = T$	$1 \in \multimap \multimap$	$\multimap \multimap \in 1$	$(RuS) \multimap = R \multimap uS \multimap$	$(RnS) \multimap = R \multimap nS \multimap$	$\multimap \multimap \multimap = \multimap$	$\multimap \multimap \multimap = \multimap$
$\circ \dashv \circ$	$(A \rightarrow B) \rightarrow (B \rightarrow A)$	$\circ \circ = 1$	$R = R \circ \circ$	$R = R \circ \circ$	$(RuS) \circ = R \circ uS \circ$	$(RnS) \circ = R \circ nS \circ$	$R \circ \circ \circ = R \circ$	$R \circ \circ \circ = R \circ$
$\multimap \triangleleft \dashv \triangleright \multimap$	$(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$	1	$(\multimap \triangleleft \otimes 1) (1 \otimes \triangleright \multimap) = 1$	$(1 \otimes \multimap \triangleleft) (\triangleright \multimap \otimes 1) = 1$	$=$	$=$	$=$	$=$
$\triangleright \multimap \dashv \multimap \triangleleft$	$(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$	1	$(1 \otimes \multimap \triangleleft) (\triangleright \multimap \otimes 1) = 1$	$(\multimap \triangleleft \otimes 1) (1 \otimes \triangleright \multimap) = 1$	$=$	$=$	$=$	$=$
$\Delta \dashv n$	$(A \rightarrow B) \rightarrow (A \rightarrow B)^2$	$R \mapsto RnR = R$	$R \in RnR$	$RnS \in R$	$\Delta(RuS) = \Delta Ru \Delta S$	$(RnT)n(SnU) = (RnS)n(TnU)$	$RnR = R$	$RnR = R$
$u \dashv \Delta$	$(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$	$(R, S) \mapsto (RuS, RuS)$	$R \in RuS$	$RuR \in R$	$(RuT)u(SuU) = (RuS)u(TuU)$	$\Delta(RnS) = \Delta Rn \Delta S$	$RuR = R$	$RuR = R$
$\perp \dashv !$	$\{*\} \rightarrow (A \rightarrow B)$	—	—	$\perp \in R$	—	—	—	—
$\cdot S \dashv /S$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	S/S	$R \in (RS)/S$	$(T/S)S \in T$	$(RuT)S = RSuTS$	$(RnT)/S = R/SnT/S$	$((RS)/S)S = RS$	$((T/S)S)/S = T/S$
$S \cdot \dashv S \setminus$	$(B \rightarrow C) \rightarrow (A \rightarrow C)$	$S \setminus S$	$R \in S \setminus (SR)$	$S(S \setminus T) \in T$	$S(RuT) = SRuST$	$S \setminus (RnT) = S \setminus RnS \setminus T$	$S(S \setminus (SR)) = SR$	$S \setminus (S(S \setminus T)) = S \setminus T$
$Rn \dashv R \Rightarrow$	$(A \rightarrow B) \rightarrow (A \rightarrow B)$	$X \mapsto R \Rightarrow (XnR)$	$X \in R \Rightarrow (XnR)$	$Rn(R \Rightarrow Y) \in Y$	$Rn(XuY) = (RnX)u(RnY)$	$R \Rightarrow (XnY) = (R \Rightarrow X)n(R \Rightarrow Y)$	$Rn(R \Rightarrow (XnR)) = XnR$	$R \Rightarrow (Rn(R \Rightarrow Y)) = R \Rightarrow Y$
$\mathcal{D} \dashv \cdot T$	$(A \rightarrow B) \rightarrow \text{Cor } A$	$R \mapsto (\mathcal{D}R)T$	$R \in (\mathcal{D}R)T$	$\mathcal{D}(AT) \in A$	$\mathcal{D}(RuS) = \mathcal{D}Ru \mathcal{D}S$	$(AnB)T = ATnBT$	$\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$	$(\mathcal{D}(AT))T = AT$
$\mathcal{R} \dashv T \cdot$	$(A \rightarrow B) \rightarrow \text{Cor } B$	$R \mapsto T(\mathcal{R}R)$	$R \in T(\mathcal{R}R)$	$\mathcal{R}(TA) \in A$	$\mathcal{R}(RuS) = \mathcal{R}Ru \mathcal{R}S$	$T(AnB) = TAnTB$	$\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$	$T(\mathcal{R}(TA)) = TA$
$\cdot f \dashv \cdot f^\circ$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	ff°	$1 \in ff^\circ$	$f^\circ f \in 1$	$(RuS)f = RfuSf$	$(RnS)f^\circ = Rf^\circ nSf^\circ$	$ff^\circ f = f$	$f^\circ ff^\circ = f^\circ$
$f^\circ \cdot \dashv f \cdot$	$(A \rightarrow C) \rightarrow (B \rightarrow C)$	ff°	$1 \in ff^\circ$	$f^\circ f \in 1$	$f^\circ(XuY) = f^\circ Xu f^\circ Y$	$f(XnY) = fXn fY$	$ff^\circ f = f$	$f^\circ ff^\circ = f^\circ$
$i \dashv E$	$\text{Map} \hookrightarrow \text{Rel}$	P	$\{\cdot\} : A \rightarrow P \quad A$	$\exists : P \rightarrow B \rightarrow B$	—	—	$\Lambda(\exists) = 1$	$\Lambda(R)\exists = R$
$\cdot \exists \dashv \Lambda$	$\text{Map}(A, P \rightarrow B) \rightarrow (A \rightarrow B)$	1	$\Lambda(f\exists) = f$	$\Lambda(R)\exists = R$	—	—	$\Lambda(f\exists)\exists = f\exists$	$\Lambda(\Lambda(R)\exists) = \Lambda(R)$
$\Lambda \dashv \cdot \exists$	$(A \rightarrow B) \rightarrow \text{Map}(A, P \rightarrow B)$	1	$\Lambda(R)\exists = R$	$\Lambda(f\exists) = f$	—	—	$\Lambda(\Lambda(R)\exists) = \Lambda(R)$	$\Lambda(f\exists)\exists = f\exists$

§1.2. Composing adjunctions

(1.2a)

$$\begin{array}{c}
\boxed{\frac{X}{f(R/T)}} \xLeftrightarrow{f^\circ \cdot \dashv f \cdot} \boxed{\frac{f^\circ X}{R/T}} \xLeftrightarrow{\cdot T \dashv /T} \boxed{\frac{(f^\circ X) T}{R}} \\
\text{f a map} \\
\\
\xLeftrightarrow{\text{associativity}} \boxed{\frac{f^\circ (X T)}{R}} \xLeftrightarrow{f^\circ \cdot \dashv f \cdot} \boxed{\frac{X T}{f R}} \xLeftrightarrow{\cdot T \dashv /T} \boxed{\frac{X}{(f R)/T}} \\
\\
\xRightarrow{\text{indirect equality}} \boxed{\frac{f(R/T)}{(f R)/T}}
\end{array}$$

	$\cdot T$	$T \cdot$	$\cdot g$	$g^\circ \cdot$	Tn	$^\circ$	$\circ \triangleleft$	Δ	(1.2b)
$\cdot S$	$R/(ST) = (R/T)/S$	$T \setminus (R/S) = (T \setminus R)/S$	$R/(Sg) = (Rg^\circ)/S$	$g(R/S) = (gR)/S$	—	$(Y/S)^\circ = S^\circ \setminus (Y^\circ)$	$(RS)^\wedge = R^\wedge(S \otimes 1)$	$(T_1 n T_2)/S = T_1/S n T_2/S$	
$S \cdot$	$S \setminus (R/T) = (S \setminus R)/T$	$(TS) \setminus R = S \setminus (T \setminus R)$	$S \setminus (Rg^\circ) = (S \setminus R)g^\circ$	$(g^\circ S) \setminus R = S \setminus (gR)$	—	$(S \setminus Y)^\circ = Y^\circ/S^\circ$	$((S \otimes 1)R)^\wedge = S \ R^\wedge$	$S \setminus (T_1 n T_2) = S \setminus T_1 n S \setminus T_2$	
$\cdot f$	$R/(fT) = (R/T)f^\circ$	$T \setminus (Rf^\circ) = (T \setminus R)f^\circ$	$(fg)^\circ = g^\circ f^\circ$	—	—	$(fY)^\circ = Y^\circ f^\circ$	—	$(T_1 n T_2)f^\circ = T_1 f^\circ n T_2 f^\circ$	
$f^\circ \cdot$	$f(R/T) = (fR)/T$	$(Tf^\circ) \setminus R = f(T \setminus R)$	—	$(fg)^\circ = g^\circ f^\circ$	—	$(fY)^\circ = Y^\circ f^\circ$	—	$f(T_1 n T_2) = fT_1 n fT_2$	
Rn	—	—	—	—	$(RnT) \Rightarrow Y = R \Rightarrow (T \Rightarrow Y)$	$(R \Rightarrow Y)^\circ = R^\circ \Rightarrow (Y^\circ)$	—	$R \Rightarrow (T_1 n T_2) = (R \Rightarrow T_1) n (R \Rightarrow T_2)$	
$^\circ$	$(Y/T)^\circ = T^\circ \setminus (Y^\circ)$	$(T \setminus Y)^\circ = Y^\circ/T^\circ$	$(gY)^\circ = Y^\circ g^\circ$	$(gY)^\circ = Y^\circ g^\circ$	$(T \Rightarrow Y)^\circ = T^\circ \Rightarrow (Y^\circ)$	$Y^\circ = Y$	$(R^\wedge)^\circ = (R^\circ \otimes 1)(1 \otimes \triangleright \circ)$ both bends: the $^\circ$ section's display	$(T_1 n T_2)^\circ = T_1^\circ n T_2^\circ$	
$\circ \triangleleft$	$(RT)^\wedge = R^\wedge(T \otimes 1)$	$((T \otimes 1)R)^\wedge = T \ R^\wedge$	—	—	—	$(R^\wedge)^\circ = (R^\circ \otimes 1)(1 \otimes \triangleright \circ)$ both bends: the $^\circ$ section's display	—	—	
u	$(X_1 u X_2)T = X_1 T u X_2 T$	$T(X_1 u X_2) = TX_1 u TX_2$	$(X_1 u X_2)g = X_1 g u X_2 g$	$g^\circ(X_1 u X_2) = g^\circ X_1 u g^\circ X_2$	$Tn(X_1 u X_2) = (TnX_1)u(TnX_2)$	$(X_1 u X_2)^\circ = X_1^\circ u X_2^\circ$	—	—	
$\perp$	$\perp T = \perp$	$T \perp = \perp$	$\perp g = \perp$	$g^\circ \perp = \perp$	$Tn\perp = \perp$	$\perp^\circ = \perp$	—	—	

§1.3. Adjoint triples

Beyond $u \dashv \Delta \dashv n$, which §1.1 already carries as two rows, the table holds one adjoint triple with content: $\exists_f \dashv f^* \dashv \forall_f$ along a map f , once on each side of the composite. $f : A \rightarrow B$ throughout this subsection.

$$\begin{aligned} \cdot f &\dashv \cdot f^\circ \dashv /f^\circ \\ f^\circ \cdot &\dashv f \cdot \dashv f \setminus \end{aligned} \quad (1.3a)$$

The first two links of each chain are the map shunting rules, and the third is §1.1's division row read at a chosen divisor: $\cdot S \dashv /S$ at $S := f^\circ$, and $S \cdot \dashv S \setminus$ at $S := f$.

The third link is not the first over again. For a map f , $/f$ collapses to $\cdot f^\circ$; being a map is exactly what that collapse spends $R/(f \ S) = (R/S) \ f^\circ$. — f° is not a map, so $/f^\circ$ does not collapse in turn, and the three operators stay distinct. On $\text{Rel}(\mathbb{I}, A) = P \ A$ they are the image triple:

operator	acts by	name	(1.3b)
$\cdot f$	$S \mapsto f[S]$	direct image, $\exists_f$	
$\cdot f^\circ$	$T \mapsto f^{-1}[T]$	inverse image, f^*	
$/f^\circ$	$S \mapsto \{b : f^{-1}(b) \subseteq S\}$	$\forall_f$	

§1.1's $\mathcal{D} \dashv \cdot T$ is this same chain along the projection $A \otimes B \rightarrow A$, read through $\text{Rel}(A, B) = P(A \otimes B)$: $\mathcal{D}R = \{(a, a) : \exists b. aRb\}$, AT is the relation that ignores b altogether, and the third link is $R \mapsto 1 \ n \ R/T = \{(a, a) : \forall b. aRb\}$.

$$\mathcal{D} \dashv \cdot T \dashv 1n\cdot/T \quad (1.3c)$$

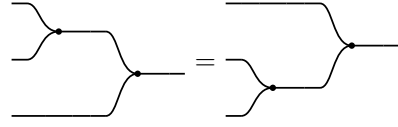
Three is where it stops, and one map breaks both ends. Take $A = \{a_1, a_2\}$ and $B = \{b\}$: $f[\{a_1\}] \ n \ f[\{a_2\}] = \{b\}$ while $f[\{a_1\} \ n \ \{a_2\}] = \emptyset$, so $\cdot f$ does not preserve meets and has no left adjoint, and the same two subsets give $\forall_f(\{a_1\} \cup \{a_2\}) = \{b\}$ against $\forall_f\{a_1\} \cup \forall_f\{a_2\} = \emptyset$, so $/f^\circ$ does not preserve joins and has no right adjoint. A monic f restores the binary case and no more — τf still reaches only $\text{im } f$, and $\forall_f \perp = B \setminus \text{im } f$ is still not $\perp$. The chain extends only when f° is a map as well, and then $/f^\circ = \cdot f^{\circ\circ} = \cdot f$ and it repeats forever. For an S with neither S nor S° a map there is no triple at all: $\cdot S \dashv /S$ is two links and stops.

The rows that chain forever say nothing by it: $^\circ$ is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and $^\circ \dashv \circ \dashv \circ$ has no end, and §1.1's other row of that kind, $\circ \triangleleft \dashv \triangleright \circ$, goes the same way.

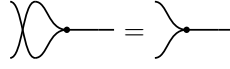
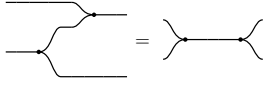
§2. Relations

$\mathbf{Rel}$ is a poset-enriched category with $(\otimes, {}^\circ)$ where $\otimes$ is commutative cartesian product, and converse ${}^\circ \dashv$, and

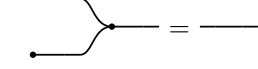
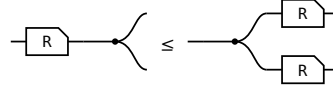
$\mathbf{C}: (\triangleright : A \otimes A \rightarrow A, \multimap : \mathbb{I} \rightarrow A)$ a commutative monoid with $\mathbf{C}^\circ \dashv \mathbf{C}$. We write $\mathbf{C}^\circ$ as $(\triangleleft : A \rightarrow A \otimes A, \multimap : A \rightarrow \mathbb{I})$



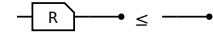
$\triangleright$ associative



$\triangleright$ commutative



$\multimap$ is its unit



Frobenius, one half — the other is its ${}^\circ$

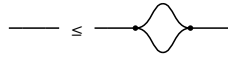
$R\triangleleft \leq \triangleleft(R\otimes R)$

$R\multimap \leq \multimap$

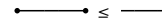
§2.1. $\forall$ object $\mathbf{A}$, $(\mathbf{A}, \triangleleft, \multimap) \dashv (\mathbf{A}, \triangleright, \multimap)$



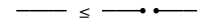
$\triangleright \triangleleft \leq 1$



$1 \leq \triangleleft \triangleright$



$\multimap \multimap \leq 1$



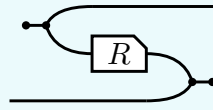
$1 \leq \multimap \multimap$

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In $\mathbf{Rel}$ it reads $\{(a, a)\} \subseteq A \times A$ — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

§3. $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

(3a)

$^\circ$ is primitive, part of the data the first section lists. It turns both of R 's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are R° , not its definition:



$$R^\circ = (\circlearrowleft \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \circlearrowright)$$

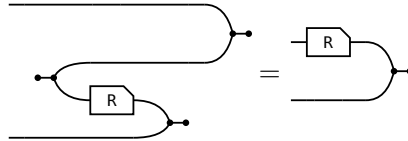
where $\circlearrowleft \triangleleft : \mathbb{I} \rightarrow a \otimes a$ opens a pair of wires out of nothing and $\triangleright \circlearrowright : b \otimes b \rightarrow \mathbb{I}$ closes one, so the input of R° is where the output of R was. Taking that formula as the definition would now be circular: $\triangleleft$ and $\circlearrowright$ are $\triangleright^\circ$ and $\circlearrowleft^\circ$. The laws of $^\circ$ are that it is a contravariant 2-functor $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$:

$$(i) \ 1^\circ = 1 \quad (ii) \ (R \ S)^\circ = S^\circ \ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§3.1. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:

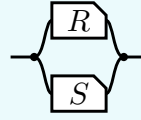
(3.1a)



R° is DEFINED as the bending of $(R \otimes 1) \triangleright \circlearrowright$, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the $\circlearrowleft \triangleleft$ bends the a strand down and the $\triangleright \circlearrowright$ brings it back up, while the b strand rides through untouched. Nothing else is spent below.

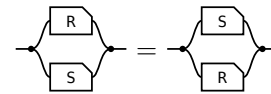
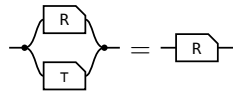
§4. $\cap$ is a commutative idempotent monoid on every hom-set

The **meet** of $R, S : a \rightarrow b$, the paper's **convolution**, is $R \cap S := \triangleleft (R \otimes S) \triangleright$ — copy the input, run R and S on the two copies, merge the results — so what comes out is what both of them do.

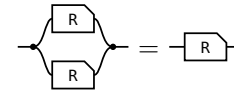
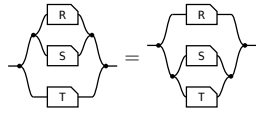


transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow $\tau = \multimap \multimap$ (the paper's Lemma 4.11).



unit: one half of τ per end — the merge's unit law absorbs the $\multimap$, the copy's counit law the $\multimap$ **commutative:** σ crosses $R \otimes S$ by naturality and is absorbed by cocommutativity and commutativity



associative: coassociativity and associativity; $\otimes$ re-brackets for nothing, **idempotent:** the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

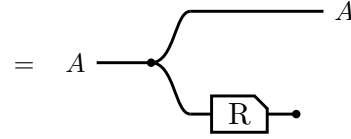
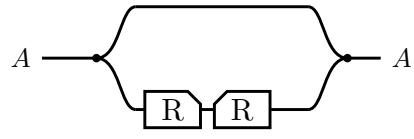
So $\leq$ is the order this monoid induces. $R \cap S \leq R$ comes from the unit, and idempotency turns anything under both S and T into something under $S \cap T$, since $R = R \cap R \leq S \cap T$.

And one law relating $\cap$ to composition, which is **not** an equation:

semi-distributivity, and what supplies it	picture
$R (S \cap T) \sqsubseteq R S \cap R T$ — the lax copy law. Equality exactly when R is single valued: the Maps section's $F (R \cap S) = F R \cap F S$.	

§5. Domain and range

The **domain** $\text{Dom } R \triangleq 1 \cap R R^\circ$ and the **range** $\text{Ran } R \triangleq \text{Dom } R^\circ$.



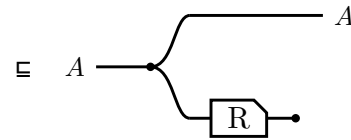
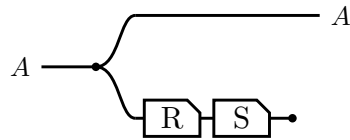
$\triangleright$ lands the return leg back on the value $\triangleleft$ handed out,
so $R^\circ \triangleright$ cuts to $\circ$ — Frobenius

Running R and throwing the result away leaves only the fact that R could fire, and $\text{Ran } R$ is the same picture with the box mirrored. In Rel both steps are $\{(a, a) : \exists b. a R b\}$.

$\text{Dom } R \triangleq 1 \cap R R^\circ$
$\text{Dom } R \subseteq A \iff R \subseteq A R, \text{ for } A \text{ coreflexive}$
$\text{Dom } (R S) \subseteq \text{Dom } R$
$\text{Dom } (R \cap S) = 1 \cap S R^\circ$
$R \text{ entire} \iff \text{Dom } R = 1 \iff 1 \subseteq R R^\circ$
$R \text{ simple} \iff R^\circ R \subseteq 1$
$R \text{ a map} \iff R \text{ entire and simple}$
$R, S \text{ entire} \implies R S \text{ entire} \text{ — likewise simple, likewise maps}$
$R S \text{ entire} \implies R \text{ entire}$

§5.1. Sliding the discard

$\text{Dom } (R S) \subseteq \text{Dom } R$, and a single glyph for Dom would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.



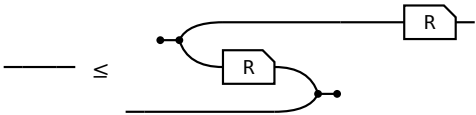
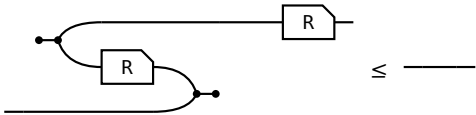
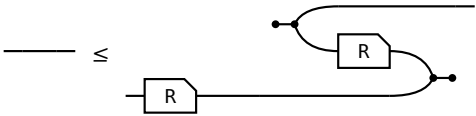
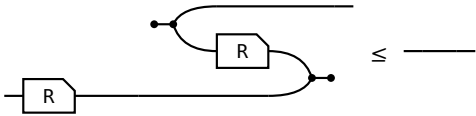
$S \circ \subseteq \circ$, the lax axiom for $\circ$ in the first section
— the discard slides back past S

Equality is $S \text{ entire}$, which is the same picture read as $\text{Dom } R = 1 \iff R \text{ entire}$.

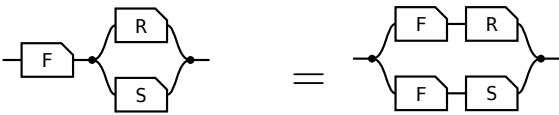
§6. Maps

Every arrow is lax for $\triangleleft$ and for $\multimap$ by axiom, and for $\triangleright$ and $\multimap$ by taking $\circ$ of those two. All four hold for **every** arrow, their REVERSES do not, and each reverse holding is a property of the arrow. The right-hand column is the **adjoint** form, which is the definition used here because it is literally the allegory's **Simple** and **Entire**; that the two forms agree is a separate theorem, not proved here.

(6a)

 surjective $1 \sqsubseteq R^\circ R$, that is, R° entire.	 single valued $R^\circ R \sqsubseteq 1$
 entire $1 \sqsubseteq R R^\circ$. With single valued, a map.	 injective $R R^\circ \sqsubseteq 1$

(6b)

law	picture
$F (R \cap S) = F R \cap F S$ F single valued F = people x may ask, R = admires a, S = admires b. Single valued means one person, so the sides agree.	 one person who admires both a at A, b at B

§7. / is all of

(7a)

$$x (R/S) y \iff \forall p. y S p \rightarrow x R p$$

$$x (S\backslash R) y \iff \forall p. p S x \rightarrow p R y$$

/ compares images: $S(y) \subseteq R(x)$. $\backslash$ compares preimages: $S^\circ(x) \subseteq R^\circ(y)$.

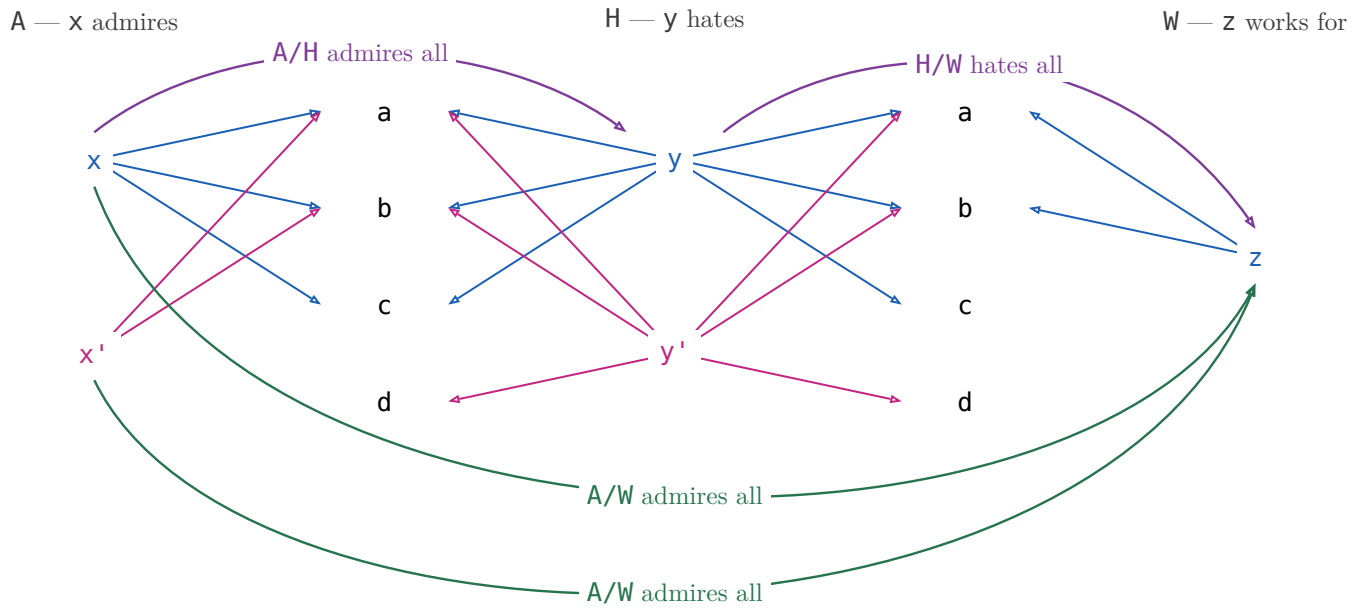
example: A admires, H hates, W works for

$x (A/H) y$ — x admires everyone y hates.

$x (H\backslash A) y$ — everyone who hates x admires y .

§7.1. $(R/S)(S/W) \sqsubseteq R/W$

(7.1a)



$$x (A/H) y \text{ — } x \text{ admires everyone } y \text{ hates}$$

$$y (H/W) z \text{ — } y \text{ hates everyone } z \text{ works for}$$

$$x (A/W) z \text{ — } x \text{ admires everyone } z \text{ works for}$$

(7.1b)

$(A/H)(H/W)$ is a path: $x \rightarrow y \rightarrow z$, and that is all of it. A/W also holds of x' , who admires everyone z works for — but x' does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of $(R/S)(S/W) \sqsubseteq R/W$.

$X \sqsubseteq R/S \iff X \ S \sqsubseteq R$ X is any X -to- Y pairing; one that only pairs X with a y such that x admires everyone y hates lies inside R/S , and R/S is the largest such.	
$X \sqsubseteq S \setminus R \iff S \ X \sqsubseteq R$ The mirror — divide on the left when X comes first.	
$(R/S) \ S \sqsubseteq R$ There is a y such that x admires everyone y hates, and p is one of the people y hates — then x admires p too. Strict at $S = \emptyset$: R/S is everyone, $(R/S) \ S = \emptyset$.	
$S \ (S \setminus R) \sqsubseteq R$ The mirror.	
associate: $R/(S_1 \ S_2) = (R/S_2)/S_1$ A friend's enemy is two hops: divide by the far end first.	
$(S_1 \ S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$ The mirror.	
maps: $f \ (R/S) = (f \ R)/S$ Rename x before or after dividing — the licence to write $f \ R / S$.	
$R/(f \ S) = (R/S) \ f^\circ$ Rename y : a map leaves a denominator as f° outside the box.	
$(R/S) \ (S/W) \sqsubseteq R/W$ Someone who admires all of a hate-set that already covers everyone Z works for admires those people too.	
$1 \sqsubseteq R/R$ R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.	
$(R/R) \ (R/R) = R/R$ R/R is the preorder admires at least as much as , and a preorder is idempotent. Freyd writes $\sqsubseteq$; with $1 \sqsubseteq R/R$ above it is an equality.	
$(R/R) \ R = R$ Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.	
$R/1 = R$ Dividing by 1 : p 's set is just $\{p\}$, so admiring all of it is admiring p .	
$R/(S_1 \cup S_2) = R/S_1 \ \cap \ R/S_2$ Admiring a combined hate-set is admiring each set in full.	
$S \setminus (R/W) = (S \setminus R)/W$ Which is why $S \setminus R/W$ needs no bracket.	

Fifteen laws, fifteen pictures, and not one shows a generator: η , υ , $\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.

§8. $\frac{R}{S}$

$$\frac{R}{S} \triangleq (R/S) \cap (S/R)^\circ. \text{ In Rel } x \text{ and } y \text{ has the same image: } \forall p. (x R p \iff y S p)$$

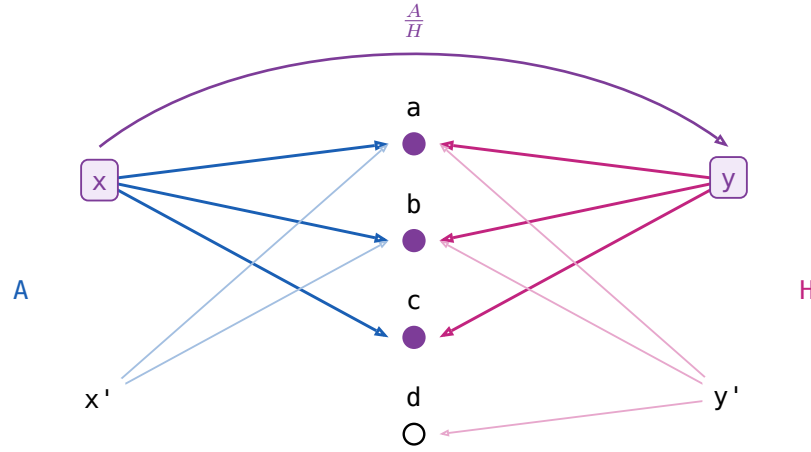
(8a)

$$\begin{aligned} x (A/H) y &\text{--- } x \text{ admires everyone } y \text{ hates} \\ x ((H/A)^\circ) y &\text{--- } x \text{ admires only people } y \text{ hates} \\ x ((A/H) \cap (H/A)^\circ) y &\text{--- } x \text{ admires only and all whom } y \text{ hates} \end{aligned}$$

(8b)

x admires exactly whom y hates: $/$ is **all of**, $\frac{R}{S}$ is **only all of**.

(8c)



law	the reading
$X \sqsubseteq \frac{R}{S} \iff XS \sqsubseteq R \text{ and } X^\circ R \sqsubseteq S$	X may pair x with y only when x admires exactly whom y hates. Both halves must typecheck, so the operation is partial .
$(\frac{R}{S})^\circ = \frac{S}{R}$	Matching is symmetric.
$\frac{R}{S} \frac{S}{W} \sqsubseteq \frac{R}{W}$	And transitive.
$\frac{R}{S} S \sqsubseteq R$	$(\exists y. x(\frac{R}{S})y \wedge ySp) \rightarrow xRp$ x only admires whom y hates $\frac{R}{S} S = \text{Dom}(\frac{R}{S})R$
$\frac{R}{R} R = R$	$(\exists y. x(\frac{R}{R})y \wedge yRp) \iff xRp$ x and y admire the same people $y = x$ always qualifies ($1 \sqsubseteq R\%R$ below)
$1 \sqsubseteq \frac{R}{R}$	$x(\frac{R}{R})y$ if x and y admires the same peoples.
$(\frac{R}{R})^2 = \frac{R}{R}$	So the relation admires the same people is an equivalence relation.
$X \sqsubseteq \frac{R}{R} \iff XR \sqsubseteq R$, for symmetric X	The largest symmetric arrow that leaves R alone.
$\frac{R}{1}$ is the simple part of R	The people who admire exactly one person and nobody else. It equals R only when R is simple, unlike $R/1 = R$.
$\text{Dom } \frac{R}{S} = 1 \cap (R/S)(S/R)$	Its domain is the domain of simplicity of R .

(8d)

§9. Freyd's Power allegories

(9a)

A **power allegory** is a division allegory with one unary operation on arrows, $\exists$ **epsilon**, subject to

$$\begin{aligned} \exists_R \square &= R\square, \quad \exists_R = \exists_R \square \\ \mathbb{1} &\sqsubseteq (R/\exists_R)(\exists_R/R) & \exists_R \text{ is } \mathbf{thick} \\ (\exists_R/\exists_R) \cap (\exists_R/\exists_R)^\circ &\sqsubseteq \mathbb{1} & \exists_R \text{ is } \mathbf{straight} \end{aligned}$$

$R\square$ is R 's target, an identity arrow. For $R : A \rightarrow B$ write $\exists : P B \rightarrow B$, dropping the subscript.

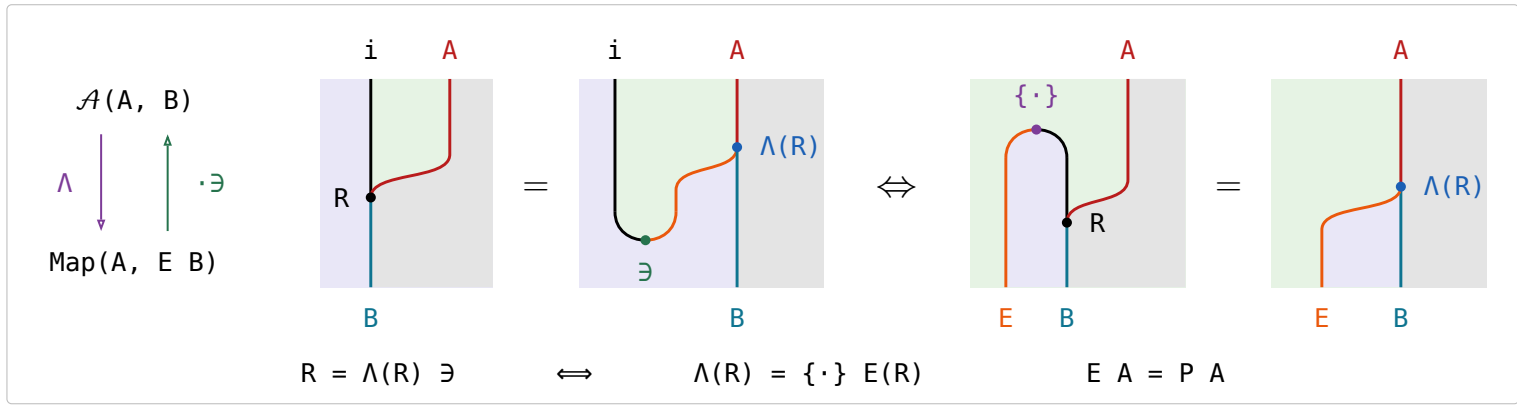
$$\epsilon \triangleq \exists^\circ : A \rightarrow P A$$

(9b)

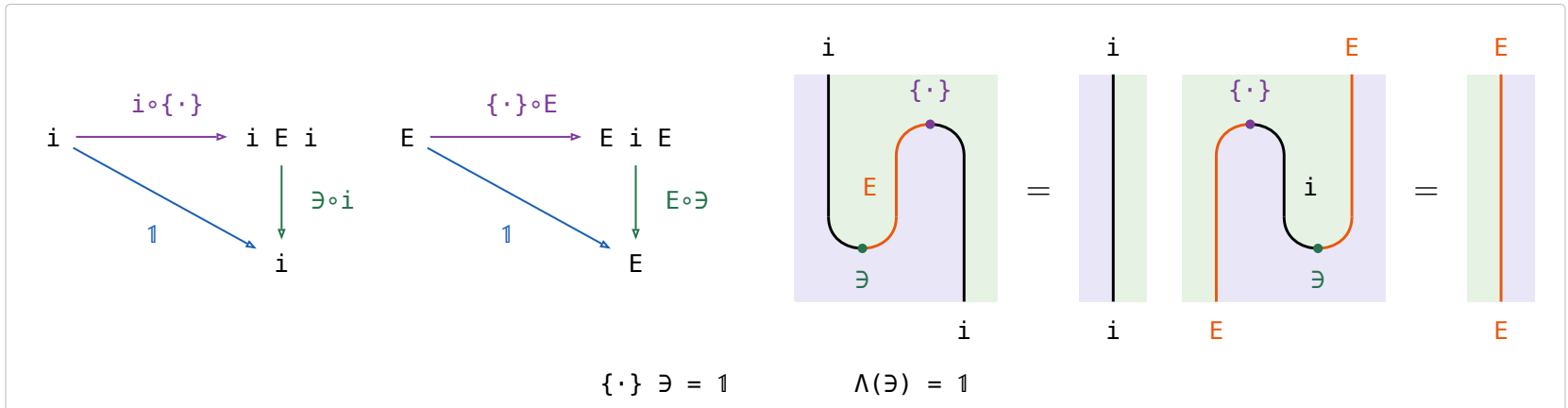
law	the reading
$\exists_R \square = R\square, \quad \exists_R = \exists_R \square$	$\exists$ has the same target as R , and replacing R by the identity at that target leaves it unchanged: one $\exists$ per object, not per arrow.
$\exists$ is thick	Comprehension: every x has a set of exactly the people x admires. Equivalently every R factors as a map followed by $\exists$.
$\exists$ is straight , that is $\frac{\exists}{\exists} \sqsubseteq \mathbb{1}$	Extensionality: two sets with the same people are the same set.
$\Lambda(R) \triangleq \frac{R}{\exists} : A \rightarrow P B$, for $R : A \rightarrow B$	convert a relation to a function. a $\Lambda(R) = \{b \mid a R b\}$, image of a
$\Lambda(R)$ is simple	$\exists$ is straight, and dividing by a straight arrow is simple. At most one set per x .
$\Lambda(R)$ is entire $\iff \exists$ thick	$\text{Dom } \frac{R}{\exists} = \mathbb{1} \cap (R/\exists)(\exists/R)$, the domain row above. At least one set per x , so $\Lambda(R)$ is a map .
$\Lambda(R) \exists = R$	Drawn in §10, where it is the counit's cancellation.
$\Lambda(R)$ is the only map with $\Lambda(R) \exists = R$	Two maps naming the same people name the same set — extensionality again.
$F \sqsubseteq \Lambda(F \exists)$, F simple	A partial choice of sets is inside the total one.
$P A \triangleq$ source of $\exists$, the power-object	Notation, not a law: $\exists$ goes $P A \rightarrow A$, and $P A$ is simply a name for where it starts. Every subset of A .
$\{\cdot\} \triangleq \Lambda(1)$, the singleton map , monic	The one-person set. $\Lambda(1)\Lambda^\circ(1) \sqsubseteq \frac{\mathbb{1}}{\exists} \frac{\exists}{\mathbb{1}} \sqsubseteq \frac{\mathbb{1}}{\mathbb{1}} \sqsubseteq \mathbb{1}$.
$\Lambda(\exists) = 1$	Make the set of a set, then read it back one level down. Straightness itself, and one of the two snake laws of §10; the other is $\{\cdot\} \exists = 1$.
fusion: $\Lambda(f R) = f \Lambda(R)$, f a map	Naturality of the unit, $f \{\cdot\} = \{\cdot\} (E(f))$, and that is the whole content of fusion. Drawn in §10.
$\Lambda(f) = f \{\cdot\}$, f a map	Rename first or take singletons first — the fusion row above at $R = 1$.
$\frac{R}{S} = \Lambda(R)\Lambda^\circ(S)$	x and y match exactly when $\Lambda(R)x$ and $\Lambda(S)y$ are one and the same set.
$E(R) \triangleq \Lambda(\exists R)$	$E(R) : P A \rightarrow P B$, image of a set of A
$\Lambda(R) = \{\cdot\} E(R)$	$\{\cdot\} : x \mapsto \{x\}$
$\Lambda(S) E(R) = \Lambda(S R)$	absorption
subset $\triangleq \exists/\exists : P A \rightarrow P A$	x subset $y \iff \forall a. y \ni a \rightarrow x \ni a$, that is $y \subseteq x$: diagram order reverses B&dM's argument order along with the arrow. It models inclusion between sets, where $\sqsubseteq$ is the allegory's own primitive comparing arrows (B&dM p. 106).

§10. $i \dashv E$ Power Allegory defined as adjunction

(10a)

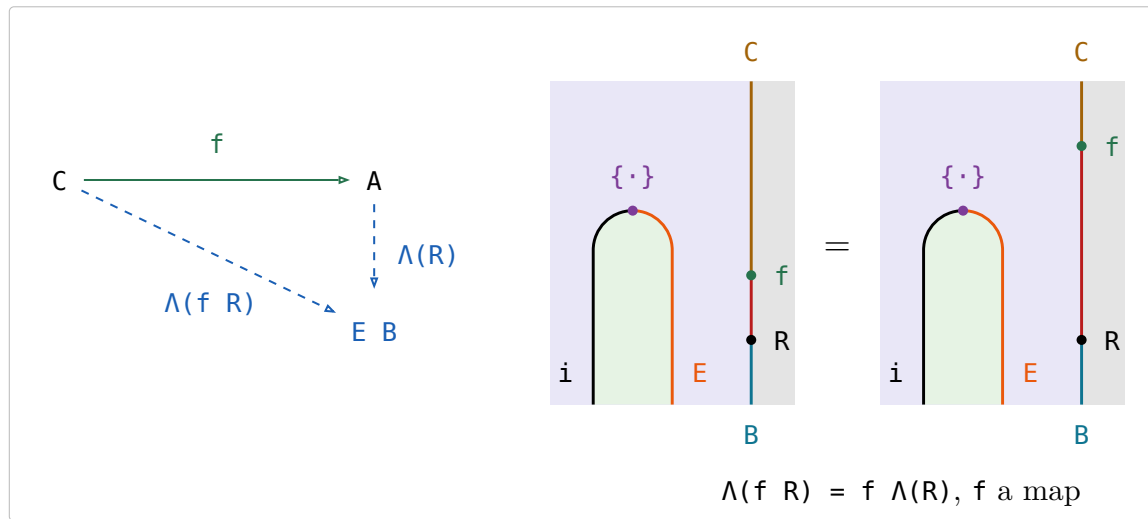


(10b)

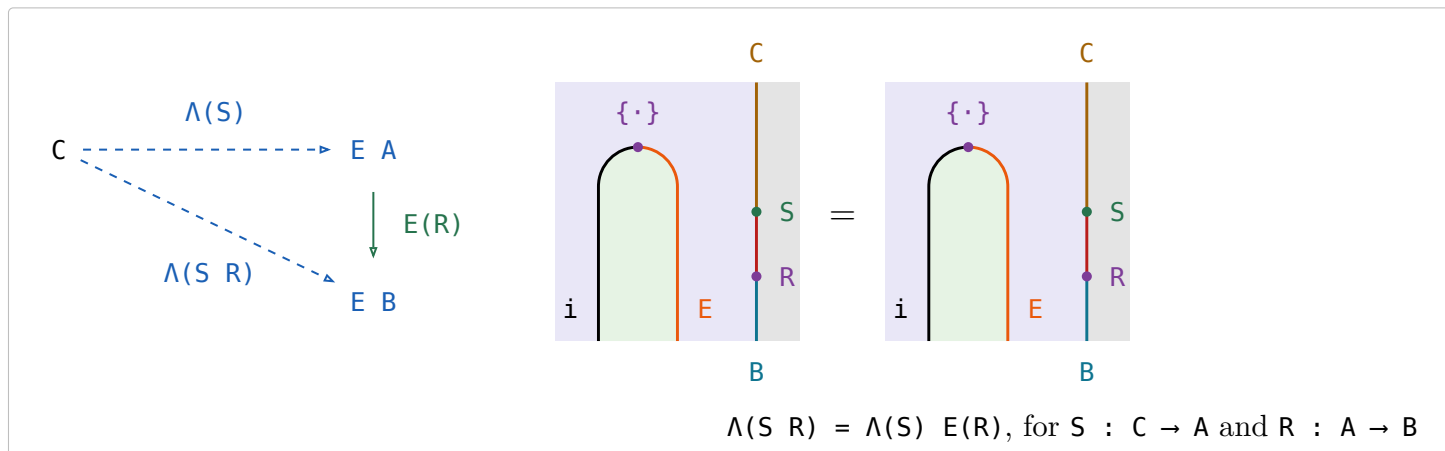


§10.1. Fusion law

(10.1a)



(10.1b)



scan line	the object on it	the bead just passed
the top edge	C	—
between $\{\cdot\}$ and S	$i \circ C$	$\{\cdot\} : C \rightarrow E C$
between S and R	$i \circ A$	S, inside the pair, so $E(S) : E C \rightarrow E A$
the bottom edge	$i \circ B$	R, likewise $E(R) : E A \rightarrow E B$

(10.1c)

One string of beads, two groupings:

- The first two, then the third: $\{\cdot\} E(S) = \Lambda(S) : C \rightarrow E A$ followed by $E(R) : E A \rightarrow E B$.
- All three at once: $\{\cdot\} E(S) E(R) = \{\cdot\} E(S R) = \Lambda(S R) : C \rightarrow E B$.

Absorption holds for every S , so fusion is its case $S := f$, plus one step: absorption there reads $\Lambda(f R) = \Lambda(f) E(R)$, and turning $\Lambda(f) E(R)$ into $f \Lambda(R)$ needs $\Lambda(f) = f \{\cdot\}$, which is (10.1a) at $R = 1$. That step is the unit's naturality, and it is all fusion adds.

§10.2. $\Lambda(R) = (R/\exists) \cap (\exists/R)^\circ$

$$\begin{array}{c}
 \boxed{\frac{g^\circ f}{\Lambda(R)}} \xleftrightarrow{\bar{f}^\circ \cdot \dashv f \cdot} \boxed{\frac{\underline{f}}{g \Lambda(R)}} \xleftrightarrow{\text{fusion}} \boxed{\frac{\underline{f}}{\Lambda(g R)}} \xleftrightarrow{\cdot \exists \dashv \Lambda} \boxed{\frac{\underline{f \exists}}{g R}} \\
 \text{f, g maps} \\
 \begin{array}{ccccc}
 & \boxed{\frac{f \exists}{g R}} & \xleftrightarrow{\bar{f}^\circ \cdot \dashv f \cdot} & \boxed{\frac{g^\circ f \exists}{R}} & \xleftrightarrow{\cdot S \dashv / S} & \boxed{\frac{g^\circ f}{R/\exists}} \\
 \xleftrightarrow{\text{antisymmetry}} & \boxed{\frac{g R}{f \exists}} & & \boxed{\frac{f^\circ g R}{\exists}} & & \boxed{\frac{f^\circ g}{\exists/R}}
 \end{array} \\
 \xleftrightarrow{\circ, \text{meet}} \boxed{\frac{g^\circ f}{(R/\exists) \cap (\exists/R)^\circ}}
 \end{array} \tag{10.2a}$$

§11. Relator

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from R to S is $R \sqsubseteq S$. A **relator** $F : \mathcal{C} \rightarrow \mathcal{D}$ is a 2-functor between allegories:

$$F(1) = 1 \quad F(R \circ S) = F(R) \circ F(S) \quad R \sqsubseteq S \implies F(R) \sqsubseteq F(S)$$

Preserving $\circ$ is **not** asked for — $\circ$ is an identity-on-objects involution $\mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$, no part of the 2-category.

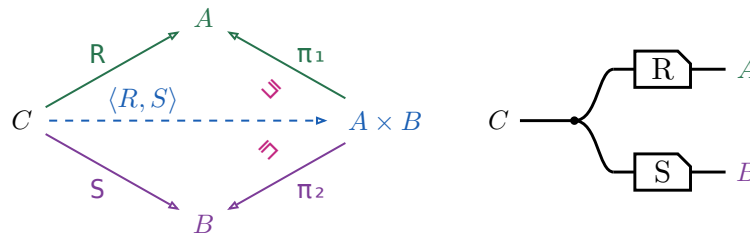
the statement	B&dM	(11b)
For f a map, $F(f)$ is a map and $F(f^\circ) = F(f)^\circ$.	Lemma 5.1	
Over a tabular allegory a functor is a relator $\iff$ it preserves $\circ$.	Theorem 5.1	
$F(R^\circ) = F(R)^\circ$ for every R , so $F(R)^\circ$ needs no bracket.	after Theorem 5.1, p. 113	
Two relators agreeing on maps are equal.	Corollary 5.1	
$F(X \cap Y) = F(X) \cap F(Y)$ for X, Y coreflexive.	Ex 5.2	
$F(R \cap S) \sqsubseteq F(R) \cap F(S)$, and strictly.	Ex 5.2, the restriction	
$F(\text{Dom } R) = \text{Dom}(F(R))$ for F preserving $\circ$.	—	

The **power relator** $P : x \mapsto P(R) : y \iff (\forall a \in x. \exists b \in y. a R b) \wedge (\forall b \in y. \exists a \in x. a R b)$ — is where the fourth is strict: for $R = \{(a_1, b_1), (a_2, b_2)\}$ and $S = \{(a_1, b_2), (a_2, b_1)\}$ the pair $(\{a_1, a_2\}, \{b_1, b_2\})$ is in $P(R) \cap P(S)$, while $R \cap S = \emptyset$.

§12. Fork $\langle R, S \rangle$

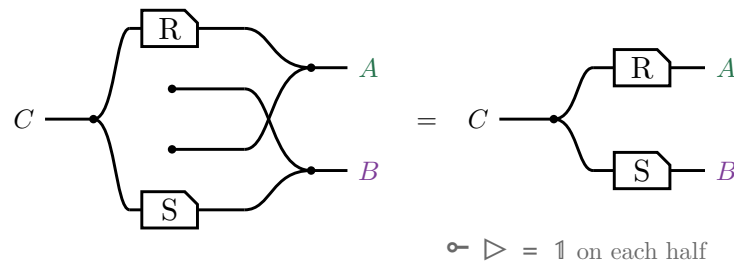
The **fork** of $R : C \rightarrow A$ and $S : C \rightarrow B$ is $\langle R, S \rangle \triangleq R \pi_1^\circ \cap S \pi_2^\circ$, where (π_1, π_2) is the tabulation of τ .

$$\langle R, S \rangle \pi_1 = (\text{Dom } S) R \quad \langle R, S \rangle \pi_2 = (\text{Dom } R) S$$



A domain is coreflexive, so $\langle R, S \rangle \pi_1 \sqsubseteq R$, with equality exactly when S is entire; for maps both triangles commute and $\langle f, g \rangle$ is unique. In $\mathbf{Rel}$, $c \in \langle R, S \rangle(a, b)$ iff $c R a$ and $c S b$ — copy c , then R on one strand and S on the other, which is $\triangleleft (R \otimes S)$ on the right.

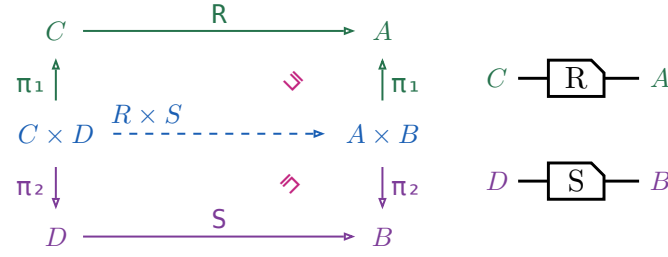
No $\circ$ survives the translation. $\pi_1 = 1 \otimes \multimap$ discards the second component, so $\pi_1^\circ = 1 \otimes \multimap$ **creates** one out of nothing, and $\cap$ is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:



§12.1. Relational product $R \times S$

$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle$, a relator in each argument but no longer a categorical product.

(12.1b)

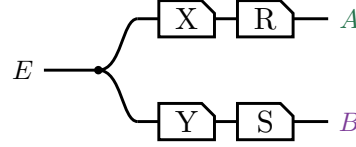


Right-then-up is $(R \times S) \pi_1$, up-then-right is $\pi_1 R$, and $(R \times S) \pi_1 \sqsubseteq \pi_1 R$, equality when S is entire. In $\mathbf{Rel}$, $(c, d) (R \times S) (a, b)$ iff $c R a$ and $d S b$ — two strands side by side, no copy dot: $R \times S = R \otimes S$.

§12.2. Absorption

For $X : E \rightarrow C$ and $Y : E \rightarrow D$, $\langle X, Y \rangle (R \times S) = \langle X R, Y S \rangle$. Both sides are this picture:

(12.2a)



§13. Coproduct $[R, S]$

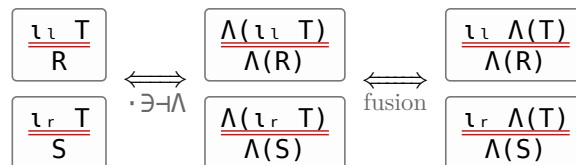
(13a)

the statement	picture
$[R, S] \triangleq \iota_l^\circ R \cup \iota_r^\circ S$ The tape is the union — a particle entering at $A + B$ takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint.	
$[R, S] = [\Lambda(R), \Lambda(S)] \ni$	
$R + S \triangleq [R \iota_l, S \iota_r]$	
$\iota_l [R, S] = R$, $\iota_r [R, S] = S$, and $[R, S]$ is the only such arrow	
$\iota_l \iota_l^\circ = \mathbb{1} = \iota_r \iota_r^\circ$	
$\iota_l \iota_r^\circ = \perp = \iota_r \iota_l^\circ$	
$\iota_l^\circ \iota_l \cup \iota_r^\circ \iota_r = \mathbb{1}$	
$[U, V]^\circ [R, S] = U^\circ R \cup V^\circ S$	

§13.1. $[R, S] \triangleq [\Lambda(R), \Lambda(S)] \ni$

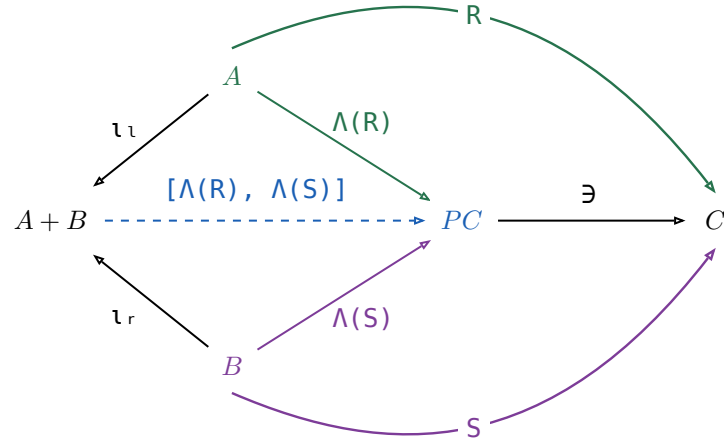
The universal-property row is not free: ι_l , ι_r were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because Λ sends an arrow $A \rightarrow C$ to a map $A \rightarrow P C$ reversibly, so the map coproduct can be applied underneath it. For any $T : A + B \rightarrow C$,

(13.1a)



$$\begin{array}{c} \longleftrightarrow \\ \text{coproduct of maps} \end{array} \boxed{\frac{\Lambda(T)}{[\Lambda(R), \Lambda(S)]}} \begin{array}{c} \longleftrightarrow \\ \cdot \exists \dashv \Lambda \end{array} \boxed{\frac{T}{[\Lambda(R), \Lambda(S)] \exists}}$$

(13.1b)



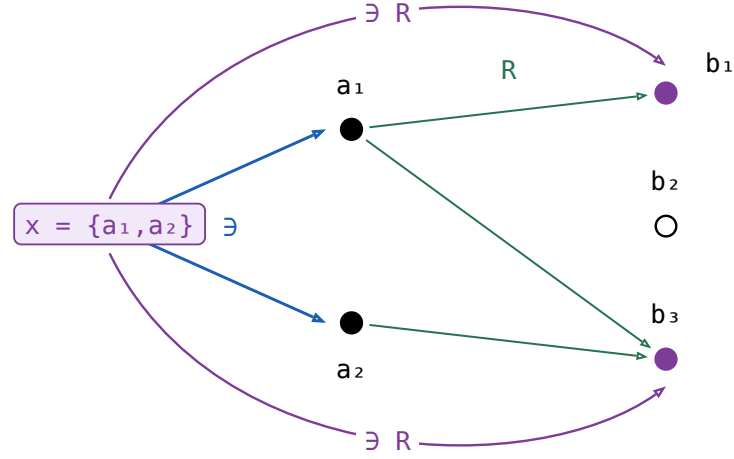
Nothing here holds only up to Ξ : every triangle commutes on the nose, which is the difference from the fork above. The border spells $[R, S] = [\Lambda(R), \Lambda(S)] \exists$, and pushing $\exists$ into the union that $[\cdot, \cdot]$ on maps already is turns that back into the definition, $(\iota_l \circ \Lambda(R) \cup \iota_r \circ \Lambda(S)) \exists = \iota_l \circ R \cup \iota_r \circ S$.

§14. The power relator $P(R)$

For $R : A \rightarrow B$, $P(R) \triangleq ((\exists R)/\exists) \cap ((\exists R^\circ)/\exists)^\circ : P A \rightarrow P B$.

$$x P(R) y \iff (\forall a \in x. \exists b \in y. a R b) \wedge (\forall b \in y. \exists a \in x. a R b)$$

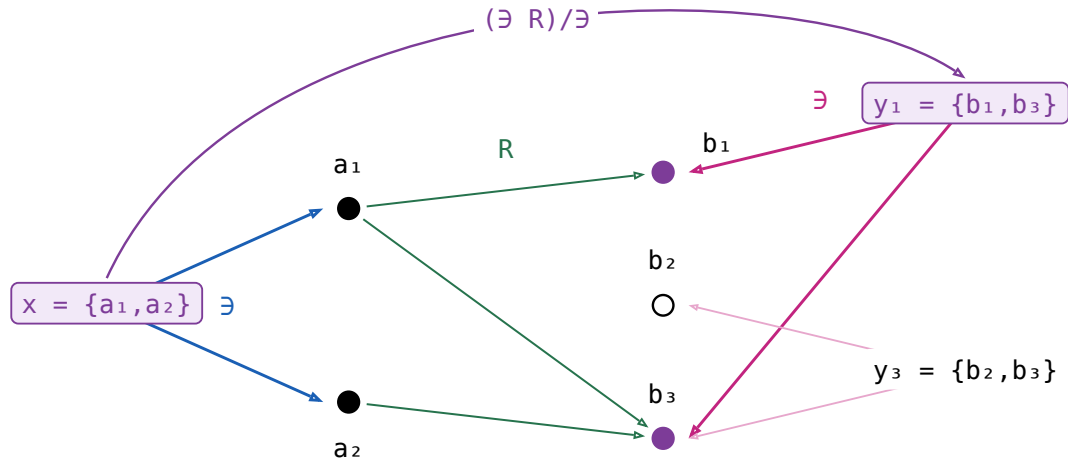
Every element of x is related by R to some element of y , and conversely.



$$\exists R : P A \rightarrow B \quad x (\exists R) b \iff \exists a \in x. a R b$$

one arc per element of B that x reaches, and b_2 gets none

Dividing by $\exists$ turns that into a relation between **sets**.

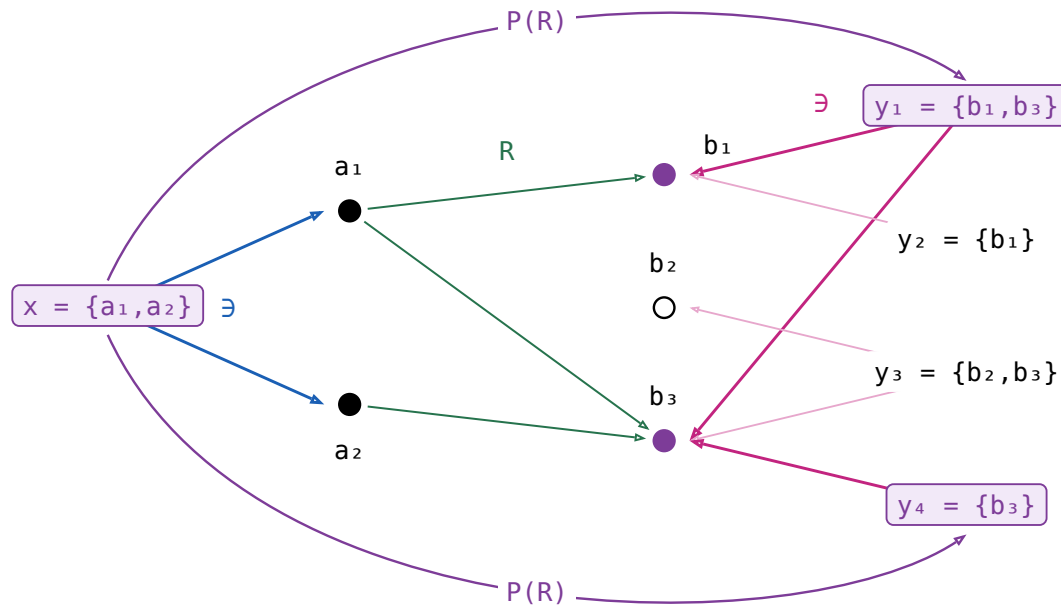


$$(\exists R)/\exists : P A \rightarrow P B \quad x ((\exists R)/\exists) y \iff \forall b \in y. \exists a \in x. a R b$$

y_3 is rejected: it names b_2 , which x does not reach

The other half of the meet is that clause with the sets swapped and R turned round: every element of x must reach y :

(14d)



$$((\exists R^\circ)/\exists)^\circ : P A \rightarrow P B \quad x ((\exists R^\circ)/\exists)^\circ y \iff \forall a \in x. \exists b \in y. a R b$$

y_2 is rejected too: a_2 's only image is b_3 , which y_2 does not name

(14e)

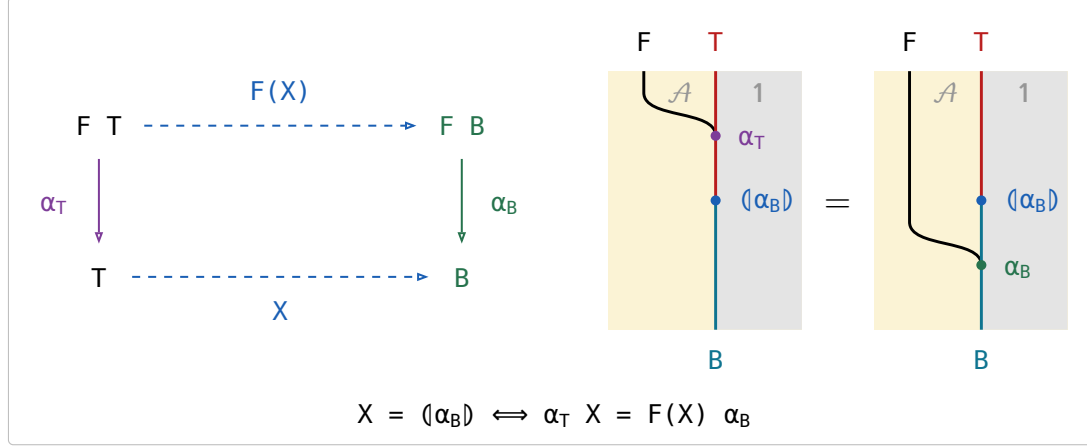
law	the reading
$X \sqsubseteq P(R) \iff X \ni \sqsubseteq \exists R$ and $X^\circ \ni \sqsubseteq \exists R^\circ$	One containment, and the same one at R° — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$, and $R \sqsubseteq S \implies P(R) \sqsubseteq P(S)$.
$P(1) = \frac{\exists}{\exists} = 1$	The straightness axiom verbatim: extensionality is P 's unit law.
$P(f) = \Lambda(\exists f) = \frac{\exists f}{\exists}$, for f a map	In $\mathbf{Rel}$, $x P(f) y \iff y = \{f a \mid a \in x\}$. The half at f° says every $a \in x$ has its $f a$ on y ; f has just the one image per a , so that already says y contains everything x reaches, which is the fraction's second half. For a map the two definitions coincide.
$P(R S) = P(R) P(S)$	$\sqsupseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(R S)$.

§15. Catamorphism

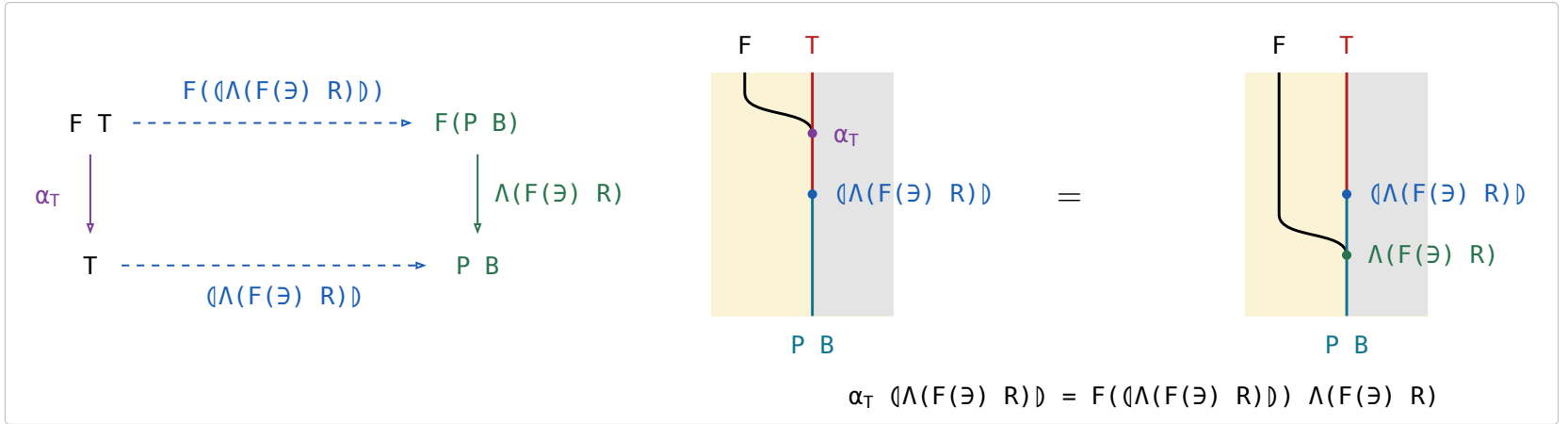
F a relator with an **initial algebra** $\alpha_T : F\ T \rightarrow T$ among the maps. For a relational algebra $\alpha_B : F\ B \rightarrow B$, the **catamorphism** $\langle \alpha_B \rangle : T \rightarrow B$ is the unique arrow with $\alpha_T \langle \alpha_B \rangle = F(\langle \alpha_B \rangle) \alpha_B$. Explicitly, for a relation $R : F\ B \rightarrow B$, $\langle R \rangle \triangleq \langle \Lambda(F(\exists)\ R) \rangle \exists$, the inner $\langle \cdot \rangle$ being the catamorphism of a **map** algebra.

An F -algebra is a **weakened** natural transformation. A transformation $F \Rightarrow 1$ on $\mathcal{A}$ would need a component $F\ X \rightarrow X$ at every object and a commuting square at every arrow, but F -Algebra only need it works on T and B .

§15.1. The defining equation

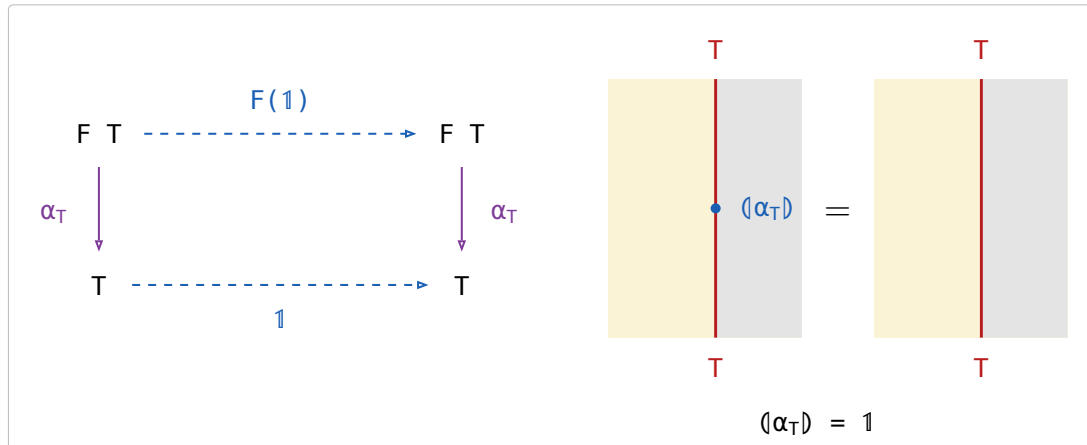


§15.2. $\langle R \rangle = \langle \Lambda(F(\exists)\ R) \rangle \exists$



$$\begin{array}{c}
 \boxed{\frac{\alpha_T X}{F(X) R}} \xleftrightarrow{\cdot \exists \dashv \Lambda} \boxed{\frac{\Lambda(\alpha_T X)}{\Lambda(F(X) R)}} \xleftrightarrow{\cdot \exists \dashv \Lambda} \boxed{\frac{\Lambda(\alpha_T X)}{\Lambda(F(\Lambda(X) \exists) R)}} \\
 \\
 \xleftrightarrow{\text{relator, fusion twice}} \boxed{\frac{\alpha_T \Lambda(X)}{F(\Lambda(X)) \Lambda(F(\exists) R)}} \xleftrightarrow{\text{catamorphism of maps}} \boxed{\frac{\Lambda(X)}{\langle \Lambda(F(\exists) R) \rangle}} \xleftrightarrow{\cdot \exists \dashv \Lambda} \boxed{\frac{X}{\langle \Lambda(F(\exists) R) \rangle \exists}}
 \end{array}
 \tag{15.2b}$$

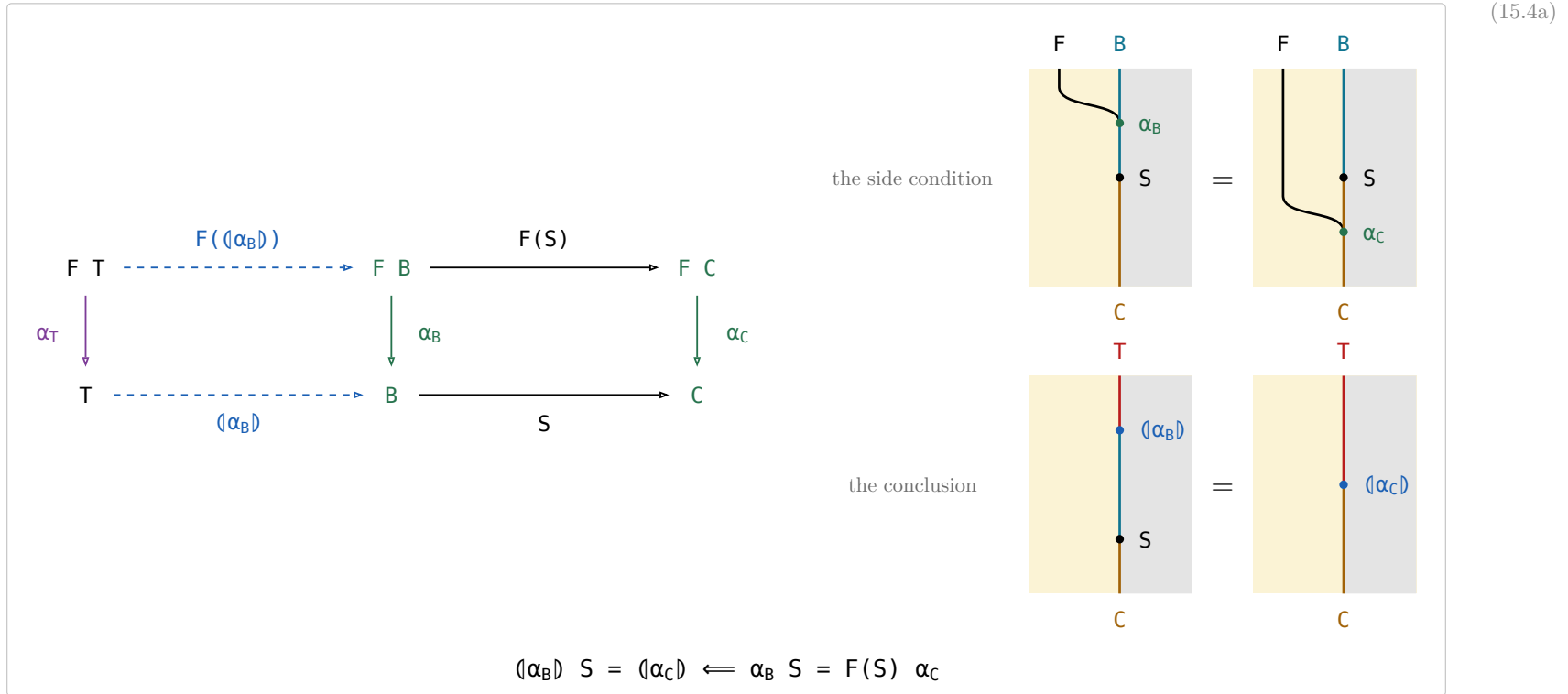
§15.3. Reflection



Taking a value apart with α_T and putting it straight back is doing nothing.

§15.4. Fusion

Fusion rewrites $(\alpha_B) S$ through a second algebra $\alpha_C : F C \rightarrow C$ along an arrow $S : B \rightarrow C$.



The left square is (α_B) 's own defining square and the right one is the side condition, so the outer rectangle says $(\alpha_B) S$ satisfies the defining equation of (α_C) — and uniqueness finishes it. A fold followed by S has collapsed into a single fold, which is how an intermediate structure is got rid of.

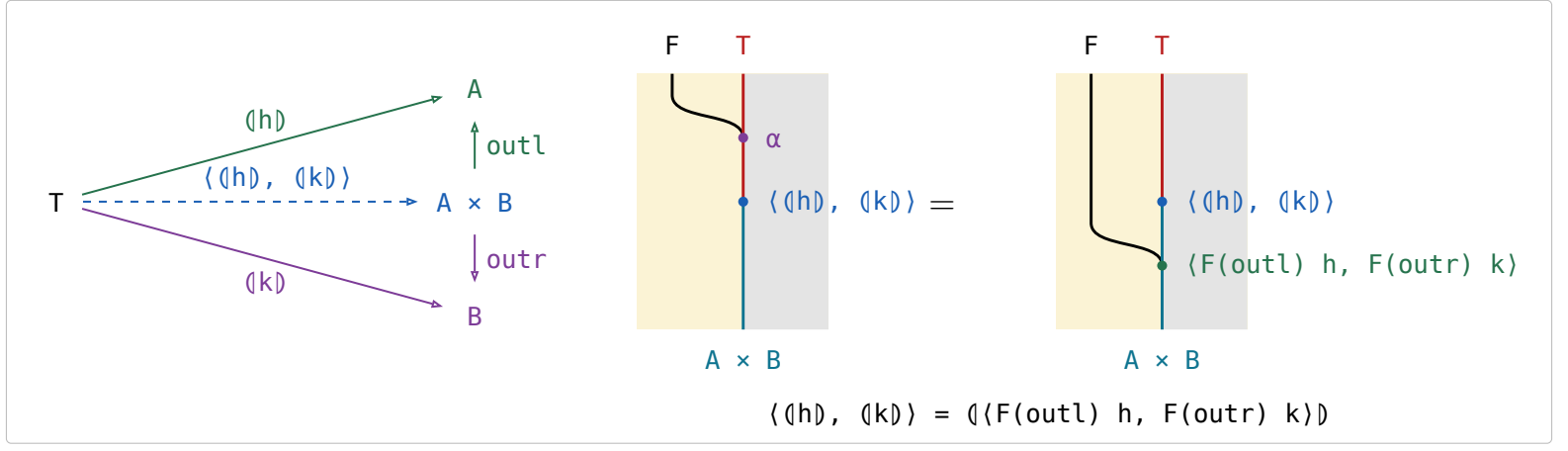
The side condition is (15.1a)'s picture again under $\alpha_T \mapsto \alpha_B$, $(\alpha_B) \mapsto S$, $\alpha_B \mapsto \alpha_C$: both say that the bead between the two merges is a homomorphism of F -algebras.

§15.5. Catamorphism in $\mathcal{S}et$

(15.5a)

name	definition	note
listr	<code>listr A ::= nil cons (A, listr A)</code>	The cons-lists over A, the datatype every row below folds (B&dM p. 55).
sum	<code>([zero, plus])</code>	<code>plus(a, b) = a + b</code> .
length	<code>([zero, outr succ])</code>	<code>outr</code> drops the head and keeps the count of the tail, <code>succ</code> adds one for the head.
average	<code>(sum, length) div</code>	<code>div(m, n) = m/n</code> , with <code>div(0, 0) = 0</code> so average is total. Traverses the list twice.
banana-split law	<code>([h], [k]) = ([F(outr) h, F(outr) k])</code>	Any fork of folds is a single fold, hence one traversal — F the base functor.
what it reduces to	<code>alpha ([h], [k]) = F([h], [k]) (F(outr) h, F(outr) k)</code>	All that (15.1a) leaves to check: the fork satisfies the defining equation, and uniqueness finishes it.
the instance	<code>(sum, length) = ([zeros, pluss])</code>	<code>zeros = (zero, zero)</code> and <code>pluss(a, (b, n)) = (a + b, n + 1)</code> , so average runs in one pass.
preds	<code>preds : Nat -> [Nat], preds n = [n, n-1, ..., 1]</code>	B&dM Ex 3.6 (p. 57), posed and not answered there: apply Ex 3.4 to write <code>preds</code> as <code>(k) outr</code> .

(15.5b)



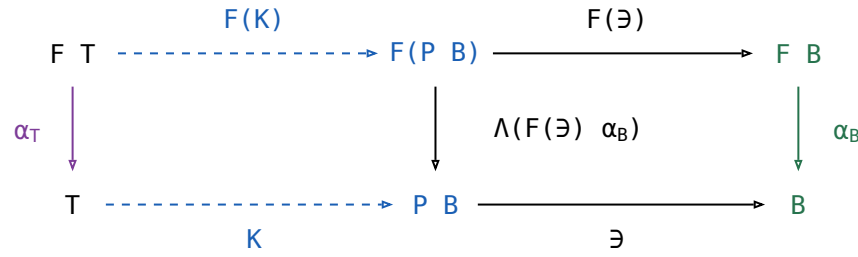
The other laws.

The two $\sqsubseteq$ **fusion** rows rewrite $\langle \alpha_B \rangle\ S$ through the same α_C and S . They are the only rows here that need the allegory **locally complete** — every hom-set a complete lattice — because they come from a least-fixed-point argument; the rest, the equality fusion (15.4a) included, needs only the initial algebra and the defining equation.

name	law	what it says
Lambek	$\alpha_T^\circ = \alpha_T^{-1}$, the initial algebra is an isomorphism	A constructor can be undone — α_T° takes a value apart into the parts it was built from.
Eilenberg–Wright	$\Lambda(\langle \alpha_B \rangle) = \langle \Lambda(F(\exists)\ \alpha_B) \rangle$	A relational fold is a deterministic fold of SETS: Λ pushes the nondeterminism into the power-object, where the fold is a map again.
Eilenberg–Wright	$\langle \alpha_B \rangle = \langle \Lambda(F(\exists)\ \alpha_B) \rangle\ \exists$	The same fact read back — fold deterministically into a set, then take a member of it.
fusion	$\langle \alpha_C \rangle \sqsubseteq \langle \alpha_B \rangle\ S \iff F(S)\ \alpha_C \sqsubseteq \alpha_B\ S$	Half of fusion: an inclusion between the two algebras is inherited by the folds.
fusion	$\langle \alpha_B \rangle\ S \sqsubseteq \langle \alpha_C \rangle \iff \alpha_B\ S \sqsubseteq F(S)\ \alpha_C$	The other half, with both inclusions turned around.

(15.5d)

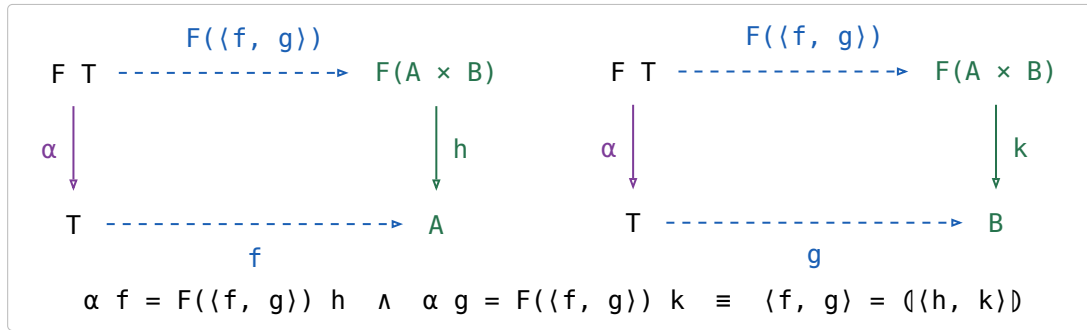
The two Λ rows, drawn:



The left square is that catamorphism's own defining square, $K = \langle \Lambda(F(\exists)\ \alpha_B) \rangle$, and the right one is Λ 's cancellation, so the outer rectangle says $K\ \exists$ satisfies the defining equation of $\langle \alpha_B \rangle$ — and uniqueness finishes it.

§15.5.1. Fokkinga's mutual recursion theorem

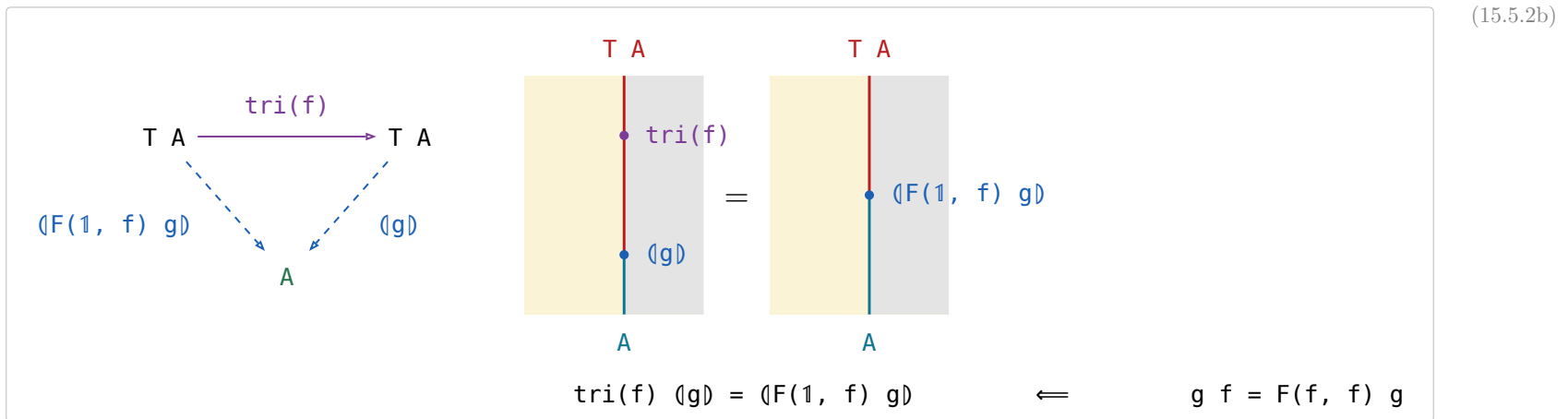
- $F\text{-Alg}(\mathcal{A})$ — the algebras and their homomorphisms — has binary products whenever $\mathcal{A}$ does, and the forgetful $U : F\text{-Alg}(\mathcal{A}) \rightarrow \mathcal{A}$ creates them: the product of $h : F A \rightarrow A$ and $k : F B \rightarrow B$ is carried by $A \times B$, with structure $\langle F(\text{outl}) h, F(\text{outr}) k \rangle : F(A \times B) \rightarrow A \times B$.
- outl and outr are homomorphisms out of it, and $\langle p, q \rangle : X \rightarrow A \times B$ is a homomorphism **iff** p and q both are — substitute the structure and compare the two forks component by component.
- The banana-split law of (15.5a) IS that product's universal property read at the initial algebra: $\langle h \rangle$ and $\langle k \rangle$ are the unique homomorphisms to (A, h) and (B, k) , so their fork is the unique homomorphism into the product, which is $\langle F(\text{outl}) h, F(\text{outr}) k \rangle$.
- Fokkinga's theorem is **strictly more general** and is not that product: in (15.5.1a) the two arrows leave $F(A \times B)$, not $F A$ and $F B$, so each sees BOTH components. The product is the case where they factor as $F(\text{outl}) h$ and $F(\text{outr}) k$; B&dM's Ex 3.4 is the other special case (p. 58). What it says: **any algebra on a product carrier is folded by a fork**.



§15.5.2. Ruby triangles

stage	definition	(15.5.2a)
informally (p. 58)	$\text{tri}(f) [a_0, a_1, \dots, a_i, \dots, a_n] = [a_0, f a_1, \dots, f^i a_i, \dots, f^n a_n]$	
for cons-lists (p. 59)	$\text{tri}(f) = \langle [\text{nil}, (1 \times \text{listr}(f)) \text{cons}] \rangle$	
with the base functor named	$F(A, B) = 1 + A \times B, \alpha = [\text{nil}, \text{cons}], \text{ so } \text{tri}(f) = \langle F(1, \text{listr}(f)) \alpha \rangle$	
abstractly (p. 59)	F a bifunctor with initial type (α, T) : $\text{tri}(f) = \langle F(1, T(f)) \alpha \rangle$	

For the definition to make sense $f : A \rightarrow A$ is required, and then $\text{tri}(f) : T A \rightarrow T A$.



(15.5.2b) is Horner's rule, generalised from cons-lists to any initial type; B&dM call it that because for cons-lists it is the schoolbook method for evaluating a polynomial (p. 58). Fusion reduces it to $F(1, T(f)) \alpha \langle g \rangle = F(1, \langle g \rangle) F(1, f) g$ (p. 59).

§15.5.3. Depth of a tree

the datatype	$\text{tree } A ::= \text{tip } A \mid \text{bin } (\text{tree } A, \text{tree } A)$, base functor $F(A, B) = A + B \times B$, initial type $([\text{tip}, \text{bin}], \text{tree})$	(15.5.3a)
the fold	$\langle [g, h] \rangle$ is the unique f with $f (\text{tip } a) = g \ a$ and $f (\text{bin } (x, y)) = h (f \ x, f \ y)$	
$\text{tree}(f)$	$\text{tree}(f) = \langle F(f, 1) \ [\text{tip}, \text{bin}] \rangle$; pointwise $\text{tree}(f) (\text{tip } a) = \text{tip } (f \ a)$ and $\text{tree}(f) (\text{bin } (x, y)) = \text{bin } (\text{tree}(f) \ x, \text{tree}(f) \ y)$	
max	$\text{max} = \langle [1, \text{bmax}] \rangle$, where $\text{bmax } (a, b)$ is the larger of a and b	
depths	$\text{depths} = \text{tree}(\text{zero}) \ \text{tri}(\text{succ})$ — replaces every tip by its depth in the tree, zero the constant function returning 0 and succ the successor	
depth	$\text{depth} = \text{depths} \ \text{max}$	

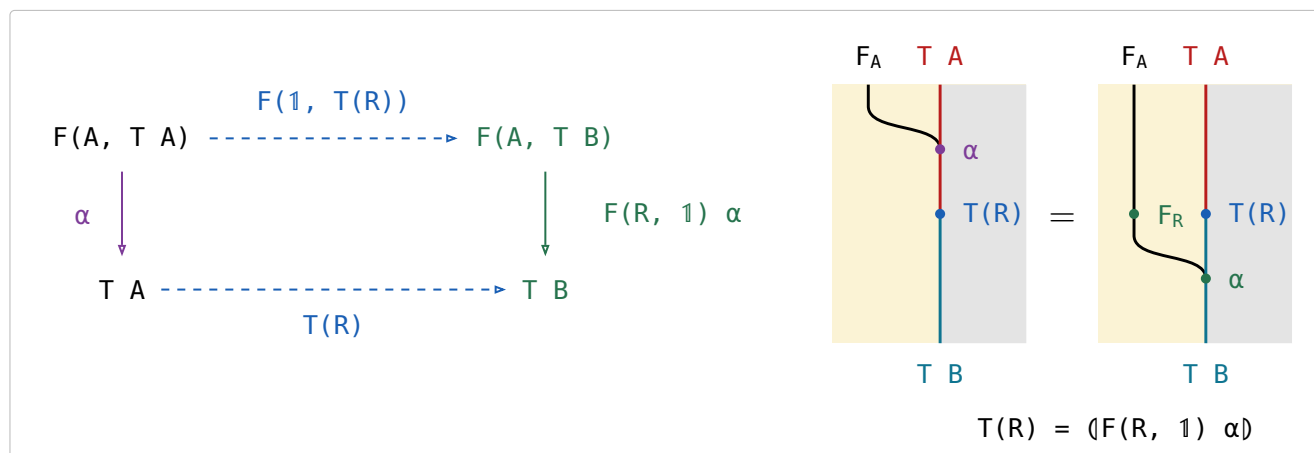
§16. Type functor

F a **binary** relator: $F(R, S)$ is its action on a pair, and $F(X)$ abbreviates $F(1, X)$, the F of the catamorphism section. For every object A the initial algebra is $\alpha : F(A, T A) \rightarrow T A$, among the maps. The **type functor** T acts on an arrow $R : A \rightarrow B$ by

$$T(R) = (F(R, 1) \alpha) : T A \rightarrow T B$$

F is the **base functor** — one layer of the structure, acting on the recursive position — while T is the datatype itself, acting on the parameter. For cons-lists, $\text{list}(R) = ([\text{nil}, (R \otimes 1) \text{cons}])$, which is $\text{map}(R)$.

A bifunctor puts its two arguments in two different places. The wire is the partial application $F_A = F(A, -)$, and an arrow in the SECOND argument is a bead on the object wire with the F wire running past it — the catamorphism section's rule, unchanged — while an arrow $R : A \rightarrow B$ in the FIRST induces a natural transformation $F_R : F_A \Rightarrow F_B$, which is a bead ON the F wire. Marsden draws bifunctors exactly so ([CatString.pdf](#), **Bifunctors**): side-by-side is already spent on functor composition, so one argument has to go into the wire's name.



$T(R)$ is the unique arrow making it commute; there is no Ξ in it.

Two beads at one height on two different wires are one action, here $F(R, T(R))$, so the right panel reads $F(R, T(R)) \alpha$ and the left one $\alpha T(R)$. Their relative height says nothing: slide F_R below the $T(R)$ bead and the panel reads $F(1, T(R)) F(R, 1)$ and then α , which is the square's top row followed by its right vertical; slide it above and the panel reads $F(R, 1) F(1, T(R))$, the route through $F(B, T A)$ the square leaves out. Identifying those two is the exchange condition every bifunctor satisfies (Hinze & Marsden, Ex 1.23), and in this calculus there is nothing to identify.

name	law	what it says
the defining equation	$T(R) = (F(R, 1) \alpha)$	Rebuild the structure with α , applying R to the parameter on the way; that is $\text{map}(R)$.
functor	$T(1) = 1$ and $T(R) T(S) = T(R S)$	Mapping the identity changes nothing, and two maps in a row are one map.
type functor fusion	$T(R) (Q) = (F(R, 1) Q)$	A map followed by a fold is a single fold — the mapped structure is never built.
naturality of α	$\alpha T(R) = F(R, T(R)) \alpha$	Building and then mapping is the same as mapping the parts and then building, so α is natural from $G(R) = F(R, T(R))$ to T .
type relator	$T(R)^\circ = T(R^\circ)$	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

Type functor fusion is the equality fusion (15.4a) applied to $T(R)$'s own defining algebra, whose side condition holds because F is a bifunctor — $F(R, 1) F(1, (Q)) = F(R, (Q)) = F(1, (Q)) F(R, 1)$ — so it too needs only the initial algebra and the defining equation.

(16d)

$$\begin{array}{ccc}
 F(A, T A) & \xrightarrow{F(R, T(R))} & F(B, T B) \\
 \downarrow \alpha & & \downarrow \alpha \\
 T A & \xrightarrow{T(R)} & T B
 \end{array}$$

It commutes strictly, and it is (16b) with the right vertical's two halves separated: push $F(R, 1)$ up into the top row, where it meets $F(1, T(R))$ as the one action $F(R, T(R))$, and α is left as the vertical. In the string diagram that is the F_R bead sliding up past $T(R)$, and what stays put is α , one bead falling past $T(R)$ from the row above to the row below — which is what “ α is natural” means.

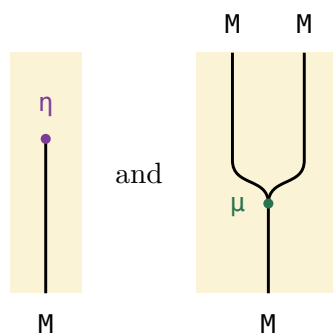
§17. Monad

(17a)

An endofunctor $M : \mathcal{A} \rightarrow \mathcal{A}$ with two natural transformations, the **unit** $\eta : \text{Id} \Rightarrow M$ and the **multiplication** $\mu : M \circ M \Rightarrow M$, subject to a unit law and an associativity law:

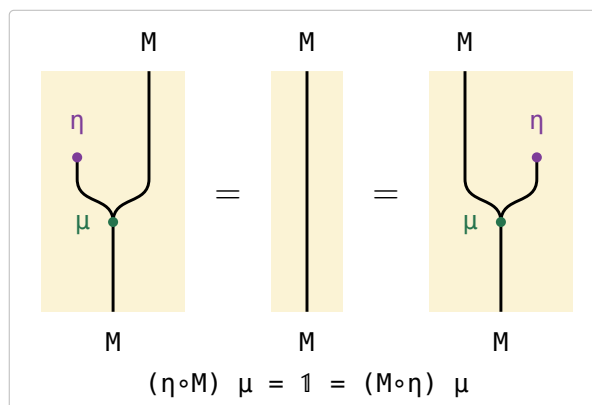
$$(\eta \circ M) \mu = 1 = (M \circ \eta) \mu \qquad (\mu \circ M) \mu = (M \circ \mu) \mu$$

(17b)



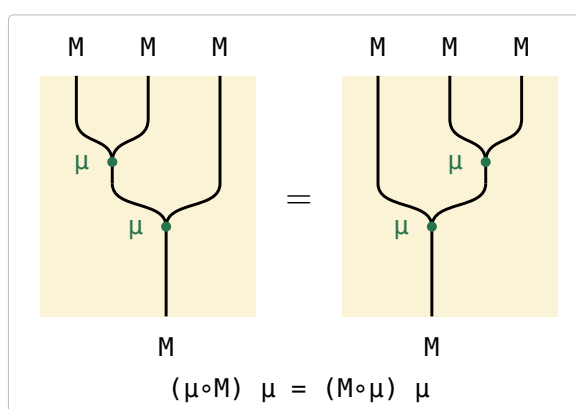
In (17b) the identity functor is a REGION and not a wire, so $\eta : \text{Id} \Rightarrow M$ has nothing above it and the M wire simply begins at its bead — the book's lollipop. $\mu : M \circ M \Rightarrow M$ is the two M wires of $M \circ M$ merging into one, its tuning fork.

(17c)



The lollipop is absorbed into the fork from either side and a bare wire is left, which is what 1 is in this calculus: nothing is drawn for it.

(17d)



The two nestings of the fork agree, so three M s collapse to one whichever pair is multiplied first.

§17.1. Monad in $\mathcal{Set}$

monad	η	μ	η on elements	μ on elements
$\text{Id } A = A$	1	1	$\eta A a = a$	$\mu A a = a$
$\text{Exc } A = A + E$	ι_l	$1 \nabla \iota_r$	$\eta A a = \iota_l a$	$\mu A (\iota_l y) = y$ $\mu A (\iota_r e) = \iota_r e$
$\text{State } A = (A \times P)^P$	$\text{curry } 1$	$(\text{apply}_{A \times P})^P$	$\eta A a = \lambda x . (a, x)$	$\mu A f = \lambda x . g y \text{ where } (g, y) = f x$
$\text{Pow } A = P A$	$\{\cdot\}$	$\bigcup$	$\eta A a = \{a\}$	$\mu A Xs = \bigcup Xs$

(17.1a)

§17.2. The state monad

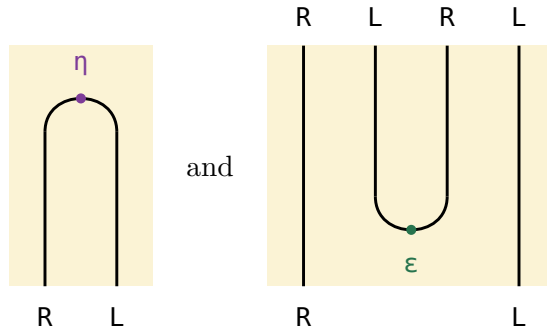
For a fixed set P of states, `IntroString p. 67` defines

$$\text{State } A = (A \times P)^P \quad \eta A = \text{curry } (1 : A \times P \rightarrow A \times P) \quad \mu A = (\text{apply}_{A \times P})^P$$

Both arrows come from the curry adjunction $L \dashv R$, $L A = A \times P$ and $R B = B^P$ (`IntroString p. 96`):

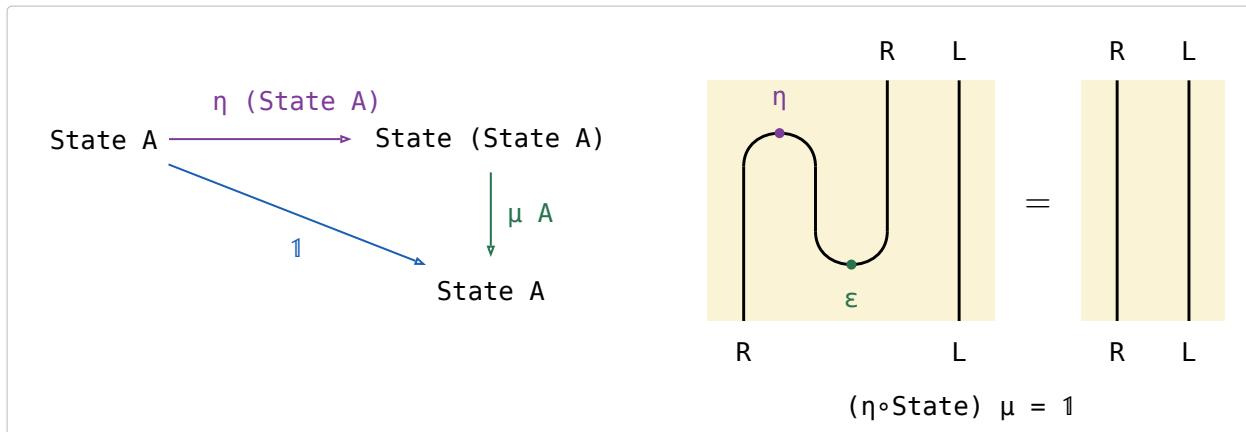
- $\text{State} = R \circ L$.
- η is the unit $\text{Id} \Rightarrow R \circ L$, which is why it is the curried identity.
- the counit $\epsilon : L \circ R \Rightarrow \text{Id}$ is `apply`.
- $\mu = R \circ \epsilon \circ L$, that counit with a functor on each side: in $\mu A = (\text{apply}_{A \times P})^P$ the subscript $A \times P$ is $L A$, the functor INSIDE, and the outer $(-)^P$ is R , the functor OUTSIDE.
- $\dashv$ composes nothing; it only names the left adjoint first.

Applicative order stands R left and L right in $\text{State} = R \circ L$. The identity functor is a region, so a unit $\text{Id} \Rightarrow R \circ L$ has nothing above it and two wires below — one strand that OPENS the pair; a counit $L \circ R \Rightarrow \text{Id}$ is that strand upside down, CLOSING a pair that arrives from above.



(17.2a)

In (17.2a), $\text{State} \circ \text{State}$ is the four wires $R L R L$, and $\mu = R \circ \epsilon \circ L$ closes the middle pair — the $L \circ R$ — leaving the outer R and L , which is State again. The two wires that run past the turn untouched are the two functors composed on, R outside and L inside.



(17.2b)

(17.2b) is the zig-zag. $\eta \circ \text{State}$ opens a new $R L$ left of the pair already there and μ closes the middle two — the new L against the old R — leaving the new R and the old L , which is State unchanged. Strip the old L , which never moves, and what zig-zags is the curry adjunction's snake equation $(\eta \circ R)(R \circ \epsilon) = 1$ (`IntroString p. 93`): (17c) read at State is that triangle identity with L composed on the inside.

In `Set` it is beta-reduction. $\eta (\text{State } A) f = \lambda x . (f, x)$ pairs the computation with the state, and μA applies it back, $\lambda x . f x$, which is f .

§18. Minimum and maximum

For $R : A \rightarrow A$, $\min R \triangleq \exists \cap (\epsilon \backslash R) : P A \rightarrow A$, $\max R \triangleq \min R^\circ$.

$$x (\min R) a \iff a \in x \wedge (\forall b \in x. b R a)$$

R points from an element to the ones at most as large, so $\min R$ picks the element of x that every element of x points to.

the law	what it says
$X \sqsubseteq \min R \iff X \sqsubseteq \exists \text{ and } \epsilon X \sqsubseteq R$	in the set, and below every element of it
$\{\cdot\} (\epsilon \backslash R) = R$	bounding a singleton is bounding its element
$\Lambda(S) (\epsilon \backslash R) = S^\circ \backslash R$	bound S 's image without building the set
$\text{union} (\epsilon \backslash R) = \epsilon \backslash (\epsilon \backslash R)$ $\text{union} \triangleq \Lambda(\exists \exists) : P(P A) \rightarrow P A$	bound a union by bounding each member set
$\{\cdot\} \min R = 1 \cap R$	a singleton's minimum is its element, where R is reflexive $\Lambda(S) \min R \text{ at } S := 1$
$\Lambda(S) \min R = S \cap (S^\circ \backslash R)$	an S -value every S -value points to
$\Lambda(S) \min R = \Lambda(S) \min(R \cap S^\circ S)$	only R between values S gives one argument counts — context
$E(S) \min R = (\exists S) \cap ((\exists S)^\circ \backslash R)$	the same for the image of a set $\Lambda(S) \min R \text{ at } S := \exists S$
$P(f) \min R = \min(f R f^\circ) f$	shunt a function through a minimum
$P(S) \min R = (\exists S) \cap (\epsilon \backslash (S R))$ R reflexive	fusion with the power relator $\exists$ is the only proof here that tabulates
$P(S) \min R \sqsubseteq (\exists S) \cap (\epsilon \backslash (S R))$	the half of the row above that costs nothing
$P(\min R) \min R \sqsubseteq \text{union} \min R$ R a preorder	a minimum in each set, then a minimum of those
$P(\min R) \min R = P(\text{Dom}(\min R)) \text{ union} \min R$ R a preorder	the same as an equality, once empty sets are dropped

§18.1. $X \sqsubseteq \min R \iff X \sqsubseteq \exists \text{ and } \epsilon X \sqsubseteq R$

$$\boxed{\frac{X}{\min R}} \xleftrightarrow{\Delta \dashv \cap} \boxed{\frac{X}{\exists} \quad \frac{X}{\epsilon \backslash R}} \xleftrightarrow{T \cdot \dashv T \backslash} \boxed{\frac{X}{\exists} \quad \frac{\epsilon X}{R}}$$

§18.2. $\{\cdot\} (\epsilon \backslash R) = R$

$$\boxed{\frac{X}{\{\cdot\} (\epsilon \backslash R)}} \xleftrightarrow{f^\circ \cdot \dashv f \cdot} \boxed{\frac{\{\cdot\}^\circ X}{\epsilon \backslash R}} \xleftrightarrow{T \cdot \dashv T \backslash} \boxed{\frac{\epsilon \{\cdot\}^\circ X}{R}} \xleftrightarrow{\circ} \boxed{\frac{(\{\cdot\} \exists)^\circ X}{R}} \xleftrightarrow{i \dashv \epsilon} \boxed{\frac{X}{R}}$$

§18.3. $\Lambda(S) (\epsilon \backslash R) = S^\circ \backslash R$

$$\boxed{\frac{X}{\Lambda(S) (\epsilon \backslash R)}} \xleftrightarrow{f^\circ \cdot \dashv f \cdot} \boxed{\frac{\Lambda(S)^\circ X}{\epsilon \backslash R}} \xleftrightarrow{T \cdot \dashv T \backslash} \boxed{\frac{\epsilon \Lambda(S)^\circ X}{R}} \xleftrightarrow{\circ} \boxed{\frac{(\Lambda(S) \exists)^\circ X}{R}} \xleftrightarrow{\cdot \exists \dashv \Lambda} \boxed{\frac{S^\circ X}{R}} \xleftrightarrow{T \cdot \dashv T \backslash} \boxed{\frac{X}{S^\circ \backslash R}}$$

§18.4. $\text{union} (\epsilon \backslash R) = \epsilon \backslash (\epsilon \backslash R)$

$$\boxed{\frac{X}{\text{union} (\epsilon \backslash R)}} \xleftrightarrow{f^\circ \cdot \dashv f \cdot} \boxed{\frac{\text{union}^\circ X}{\epsilon \backslash R}} \xleftrightarrow{T \cdot \dashv T \backslash} \boxed{\frac{\epsilon \text{union}^\circ X}{R}} \xleftrightarrow{\circ} \boxed{\frac{(\text{union} \exists)^\circ X}{R}} \xleftrightarrow{\cdot \exists \dashv \Lambda} \boxed{\frac{(\exists \exists)^\circ X}{R}} \xleftrightarrow{\circ, T \cdot \dashv T \backslash} \boxed{\frac{X}{\epsilon \backslash (\epsilon \backslash R)}}$$

§18.5. $\Lambda(S) \min R = S \cap (S^\circ \backslash R)$

$$\begin{array}{c}
 \boxed{\frac{X}{\Lambda(S) \min R}} \xLeftrightarrow{f^\circ \cdot \dashv f \cdot} \boxed{\frac{\Lambda(S)^\circ X}{\min R}} \xLeftrightarrow{\Delta \dashv n} \boxed{\frac{\Lambda(S)^\circ X}{\exists}} \xLeftrightarrow{f^\circ \cdot \dashv f \cdot} \boxed{\frac{X}{\Lambda(S) \exists}} \xLeftrightarrow{\cdot \exists \dashv \Lambda, (18.3a)} \boxed{\frac{X}{S}} \xLeftrightarrow{\Delta \dashv n} \boxed{\frac{X}{S \cap (S^\circ \setminus R)}} \\
 \boxed{\frac{\Lambda(S)^\circ X}{\exists}} \xLeftrightarrow{\cdot \exists \dashv \Lambda, (18.3a)} \boxed{\frac{X}{\Lambda(S) (\exists \setminus R)}} \xLeftrightarrow{\Delta \dashv n} \boxed{\frac{X}{S^\circ \setminus R}}
 \end{array}
 \tag{18.5a}$$

§18.6. $\Lambda(S) \min R = \Lambda(S) \min(R \cap S^\circ S)$

$$\begin{array}{c}
 \boxed{\frac{X}{\Lambda(S) \min(R \cap S^\circ S)}} \xLeftrightarrow{(18.5a)} \boxed{\frac{X}{S \cap (S^\circ \setminus (R \cap S^\circ S))}} \xLeftrightarrow{\Delta \dashv n, T \cdot \dashv T \setminus} \boxed{\frac{X}{S}} \\
 \boxed{\frac{X}{S}} \xLeftrightarrow{\Delta \dashv n, S^\circ \cdot \text{monotone}} \boxed{\frac{S^\circ X}{R}} \xLeftrightarrow{T \cdot \dashv T \setminus, \Delta \dashv n} \boxed{\frac{X}{S \cap (S^\circ \setminus R)}} \xLeftrightarrow{(18.5a)} \boxed{\frac{X}{\Lambda(S) \min R}}
 \end{array}
 \tag{18.6a}$$

§18.7. $P(f) \min R = \min(f R f^\circ) f$

$$\begin{array}{c}
 \boxed{P(f) \min R} \xLeftrightarrow{P = E \text{ on maps, (18.5a)}} \boxed{(\exists f) \cap ((\exists f)^\circ \setminus R)} \xLeftrightarrow{\cdot^\circ, T \cdot \dashv T \setminus, f^\circ \cdot \dashv f \cdot} \boxed{(\exists f) \cap (\exists \setminus (f R))} \\
 \boxed{(\exists f) \cap (\exists \setminus (f R))} \xLeftrightarrow{\text{modular law, } f \text{ simple}} \boxed{(\exists \cap ((\exists \setminus (f R)) f^\circ)) f} \xLeftrightarrow{\cdot f \dashv \cdot f^\circ, \min} \boxed{\min(f R f^\circ) f}
 \end{array}
 \tag{18.7a}$$

§18.8. $P(S) \min R = (\exists S) \cap (\exists \setminus (S R))$

(p, q) tabulates $W \triangleq (\exists S) \cap (\exists \setminus (S R))$, and $y \triangleq \Lambda((p \exists S) \cap (q R^\circ))$, a map.

$$\boxed{\frac{W}{P(S) \min R}} \xLeftrightarrow{p^\circ q = W, 1 \sqsubseteq y y^\circ} \boxed{\frac{p^\circ y}{P(S)}} \xLeftrightarrow{\cdot y^\circ q} \boxed{\frac{y^\circ q}{\min R}}
 \tag{18.8b}$$

$$\boxed{\frac{y^\circ q}{\exists}} \xLeftrightarrow{f^\circ \cdot \dashv f \cdot, \cdot \exists \dashv \Lambda} \boxed{\frac{q}{(p \exists S) \cap (q R^\circ)}} \xLeftrightarrow{\Delta \dashv n, f^\circ \cdot \dashv f \cdot} \boxed{\frac{p^\circ q}{\exists S}} \xLeftrightarrow{\Delta \dashv n, 1 \sqsubseteq R^\circ} \boxed{\frac{1}{R^\circ}}$$

R reflexive

$$\boxed{\frac{\exists y^\circ q}{R}} \xLeftrightarrow{\cdot \exists \dashv \Lambda, \cdot^\circ, \text{meet}} \boxed{\frac{R q^\circ q}{R}}$$

q simple

$$\boxed{\frac{p^\circ y}{P(S)}} \xLeftrightarrow{\Delta \dashv n, \cdot T \dashv T, \cdot^\circ, f^\circ \cdot \dashv f \cdot} \boxed{\frac{p^\circ y \exists}{\exists S}} \xLeftrightarrow{\Delta \dashv n, \cdot T \dashv T, \cdot^\circ, f^\circ \cdot \dashv f \cdot} \boxed{\frac{p \exists}{y \exists S^\circ}}$$

$$\boxed{\frac{p^\circ y \exists}{\exists S}} \xLeftrightarrow{\cdot \exists \dashv \Lambda, \text{meet}} \boxed{\frac{p^\circ p \exists S}{\exists S}}$$

p simple

$$\boxed{\frac{p \exists}{y \exists S^\circ}} \xLeftrightarrow{\text{modular law}} \boxed{\frac{p \exists}{(p \exists) \cap (q R^\circ S^\circ)}} \xLeftrightarrow{1 \sqsubseteq q q^\circ} \boxed{\frac{q^\circ p \exists}{R^\circ S^\circ}} \xLeftrightarrow{\cdot^\circ, T \cdot \dashv T \setminus} \boxed{\frac{W}{\exists \setminus (S R)}}$$

W's right half

§18.9. $P(S) \min R \sqsubseteq (\exists S) \cap (\epsilon \backslash (S R))$

(18.9a)

$$\begin{array}{ccccc}
 \boxed{\frac{P(S) \min R}{(\exists S) \cap (\epsilon \backslash (S R))}} & \xleftarrow{\Delta \dashv \cap, T \cdot \dashv T \backslash} & \boxed{\frac{P(S) \min R}{\exists S}} & \xleftarrow{\text{UP of min}} & \boxed{\frac{P(S) \exists}{\exists S}} \\
 & & \boxed{\frac{\epsilon \in P(S) \min R}{S R}} & & \boxed{\frac{\epsilon \in P(S)}{S \epsilon}}
 \end{array}$$

§18.10. $P(\min R) \min R \sqsubseteq \text{union } \min R$

(18.10a)

$$\begin{array}{ccccc}
 \boxed{\frac{P(\min R) \min R}{\text{union } \min R}} & \xleftarrow{(18.5a)} & \boxed{\frac{P(\min R) \min R}{(\exists \exists) \cap ((\exists \exists)^\circ \backslash R)}} & \xleftarrow{\circ, \Delta \dashv \cap, T \cdot \dashv T \backslash} & \boxed{\frac{P(\min R) \min R}{\exists \exists}} \\
 \text{R a preorder} & & & & \boxed{\frac{\epsilon \in P(\min R) \min R}{R}} \\
 & & & & \xleftarrow{\text{UP of min, R transitive}} \boxed{\frac{P(\min R) \exists}{\exists \min R}} \\
 & & & & \boxed{\frac{\epsilon \in P(\min R)}{\min R \epsilon}}
 \end{array}$$

§18.11. $P(\min R) \min R = P(\text{Dom } (\min R)) \text{ union } \min R$

(18.11a)

$$\begin{array}{ccc}
 \boxed{P(\min R) \min R} & \xrightarrow{\text{Dom } (\min R) \min R = \min R} & \boxed{P(\text{Dom } (\min R) \min R) \min R} \\
 \xrightarrow{\text{P a relator}} & \boxed{P(\text{Dom } (\min R)) P(\min R) \min R} & \xleftarrow{(18.10a)} \boxed{P(\text{Dom } (\min R)) \text{ union } \min R}
 \end{array}$$

(18.11b)

$$\begin{array}{ccc}
 \boxed{\frac{P(\text{Dom } (\min R)) \text{ union } \min R}{P(\min R) \min R}} & \xleftarrow{(18.8b) \text{ at } S := \min R, \Delta \dashv \cap, T \cdot \dashv T \backslash} & \boxed{\frac{P(\text{Dom } (\min R)) \text{ union } \min R}{\exists \min R}} \\
 & & \boxed{\frac{\epsilon \in P(\text{Dom } (\min R)) \text{ union } \min R}{\min R R}}
 \end{array}$$

$$\begin{array}{ccc}
 \boxed{\frac{P(\text{Dom } (\min R)) \text{ union } \min R}{\exists \min R}} & \xleftarrow{P(\text{Dom } (\min R)) \sqsubseteq 1} & \boxed{\frac{\text{union } \min R}{\exists \min R}} \\
 \xleftarrow{T(U \cap V) \sqsubseteq T U \cap T V, \cdot \exists \dashv \wedge, (18.4a)} & \boxed{\frac{(\exists \exists) \cap (\epsilon \backslash (\epsilon \backslash R))}{\exists \min R}} & \xleftarrow{\text{modular law}} \boxed{\frac{\exists (\exists \cap (\epsilon \backslash (\epsilon \backslash R)))}{\exists (\exists \cap (\epsilon \backslash R))}} \\
 & & \text{counit of } T \cdot \dashv T \backslash
 \end{array}$$

$$\boxed{\frac{\epsilon \in P(\text{Dom } (\min R)) \text{ union } \min R}{\min R R}} \xleftarrow{\epsilon \text{ lax natural}} \boxed{\frac{\text{Dom } (\min R) \epsilon \text{ union } \min R}{\min R R}}$$

$$\begin{array}{ccc}
 \xleftarrow{\text{Dom } (\min R) \sqsubseteq \min R (\min R)^\circ} & \boxed{\frac{\min R (\min R)^\circ \epsilon \text{ union } \min R}{\min R R}} & \xleftarrow{(\min R)^\circ \sqsubseteq \epsilon, \epsilon \in \text{union } \sqsubseteq \epsilon} \boxed{\frac{\min R \epsilon \min R}{\min R R}} \\
 & & \text{UP of min}
 \end{array}$$